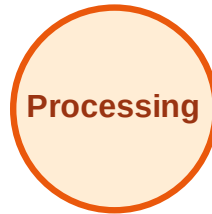


BFS Traversal

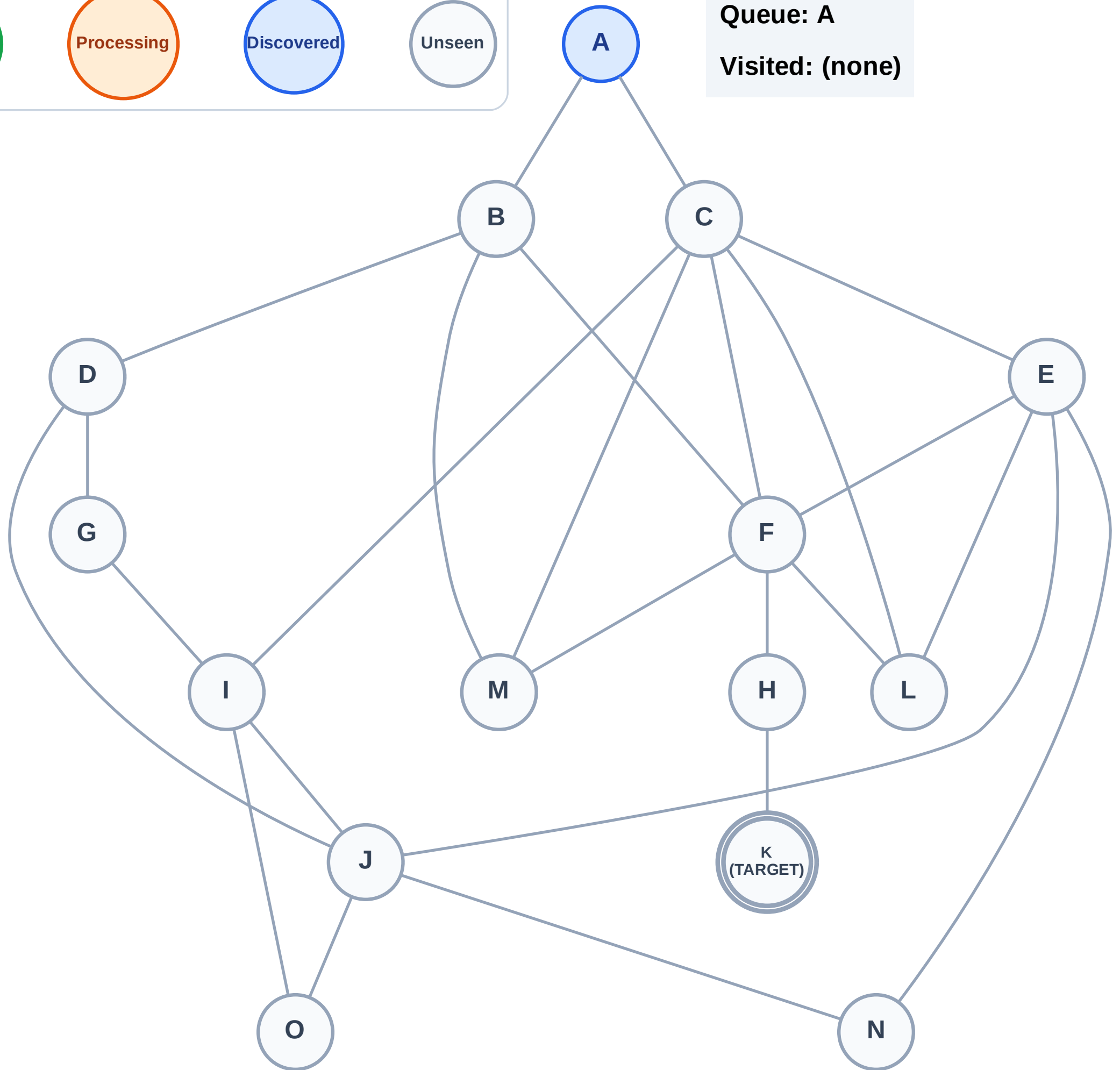
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

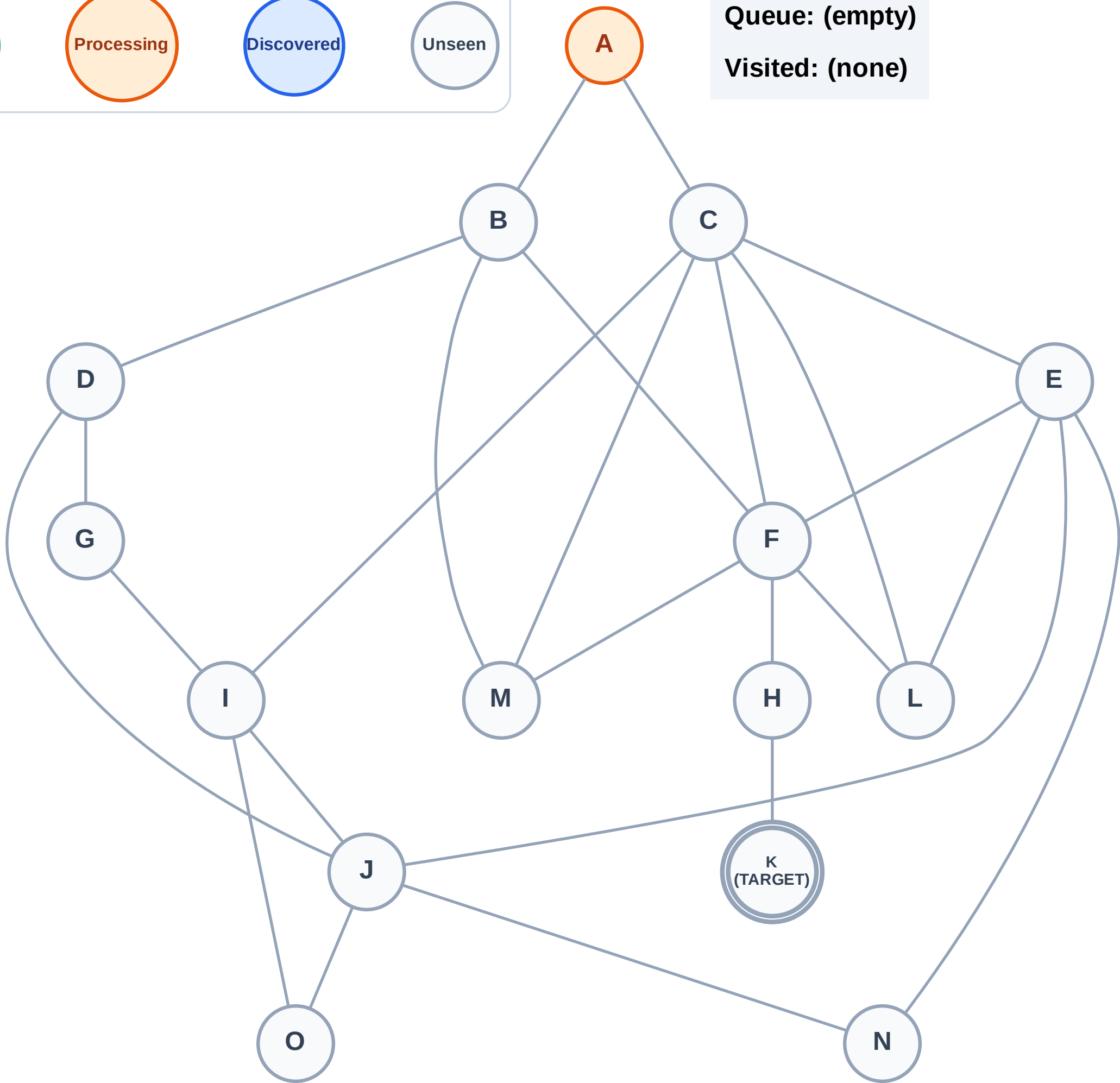
Complete

Processing

Discovered

Unseen

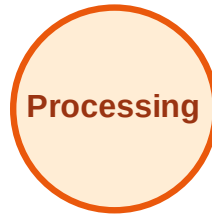
Queue: (empty)
Visited: (none)



BFS Traversal

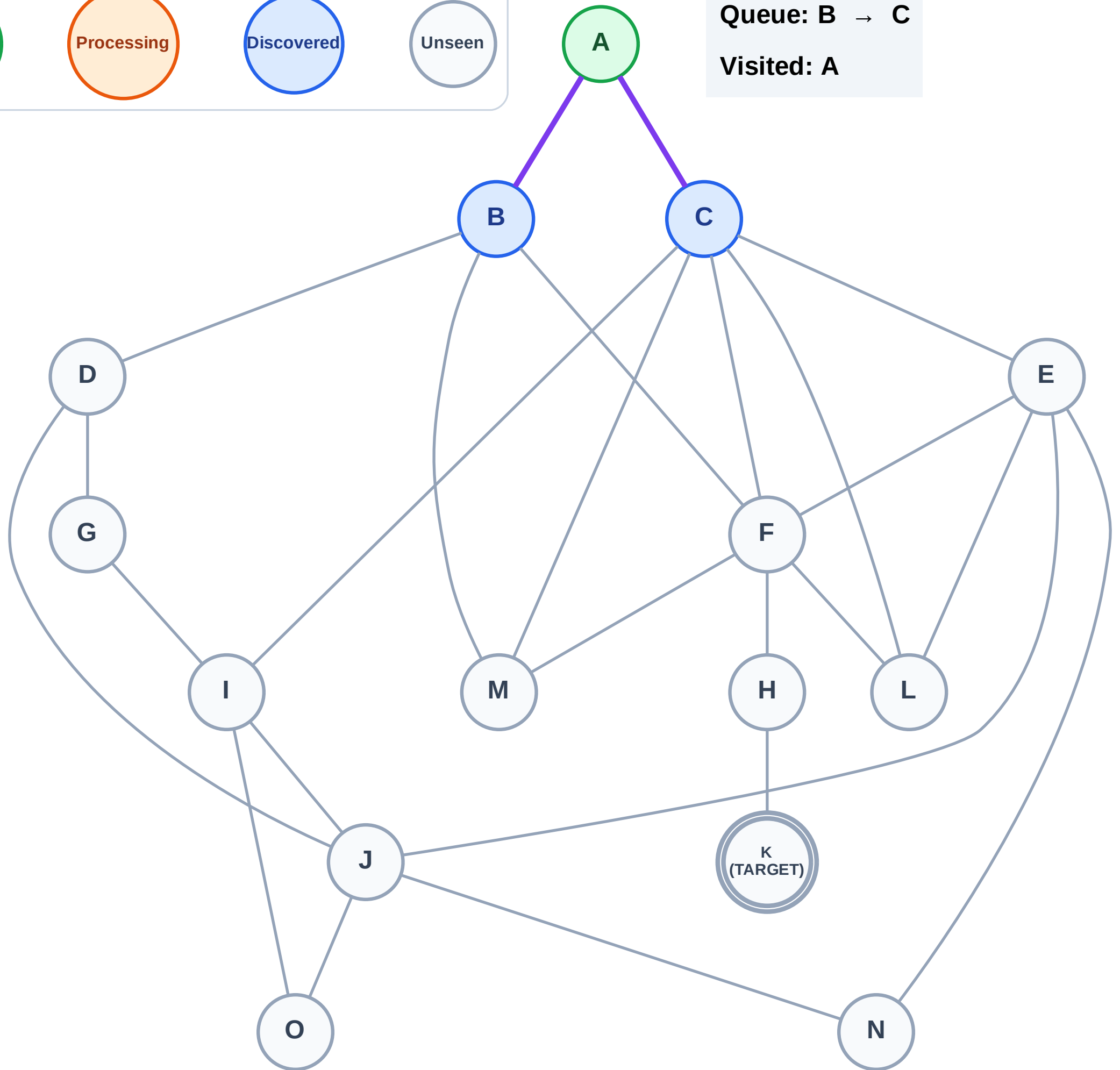
Step 3: Discover B, C from A

Legend



Queue: B → C

Visited: A



BFS Traversal

Step 4: Process node B

Legend

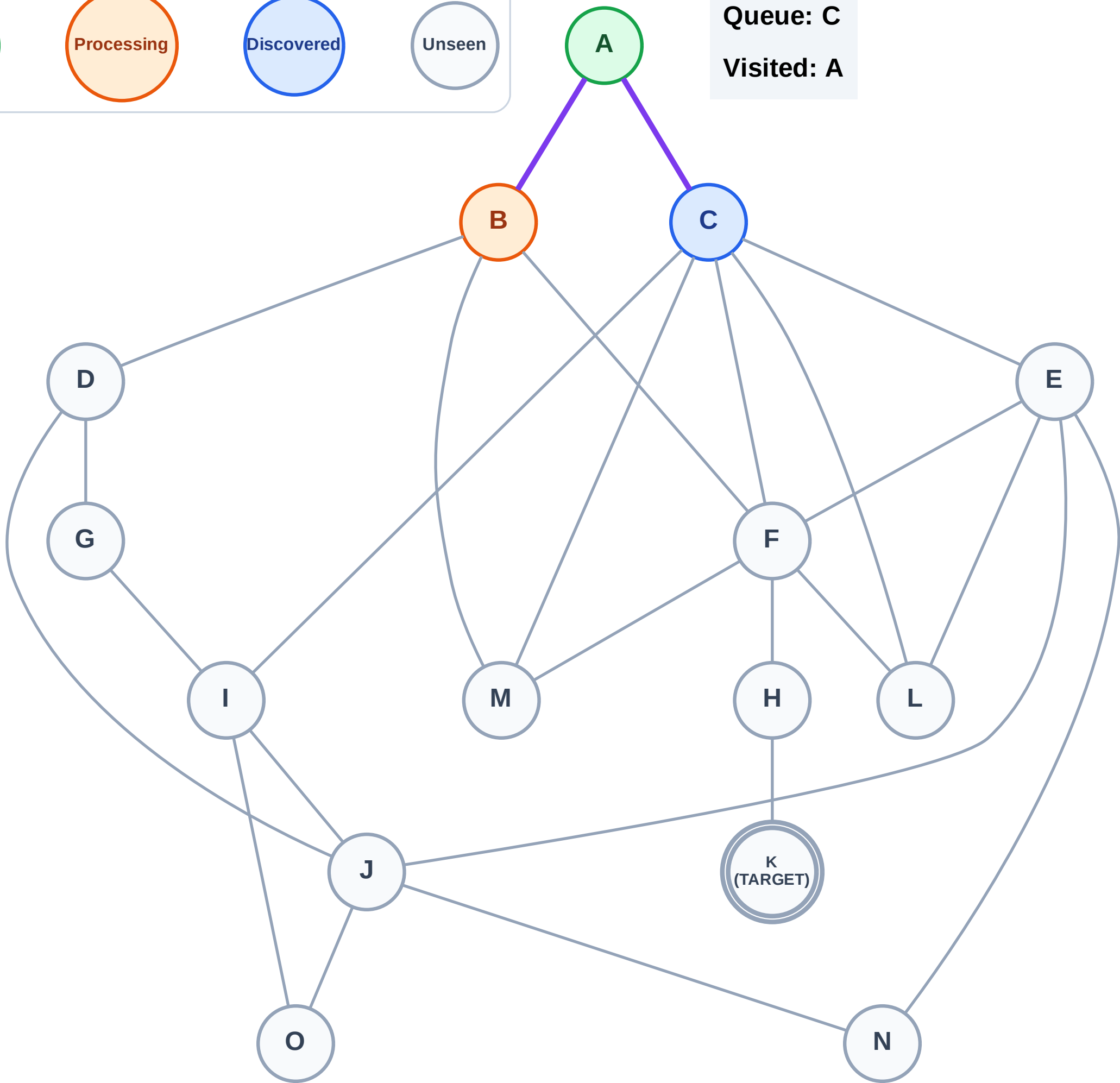
Complete

Processing

Discovered

Unseen

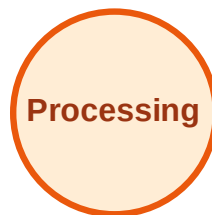
Queue: C
Visited: A



BFS Traversal

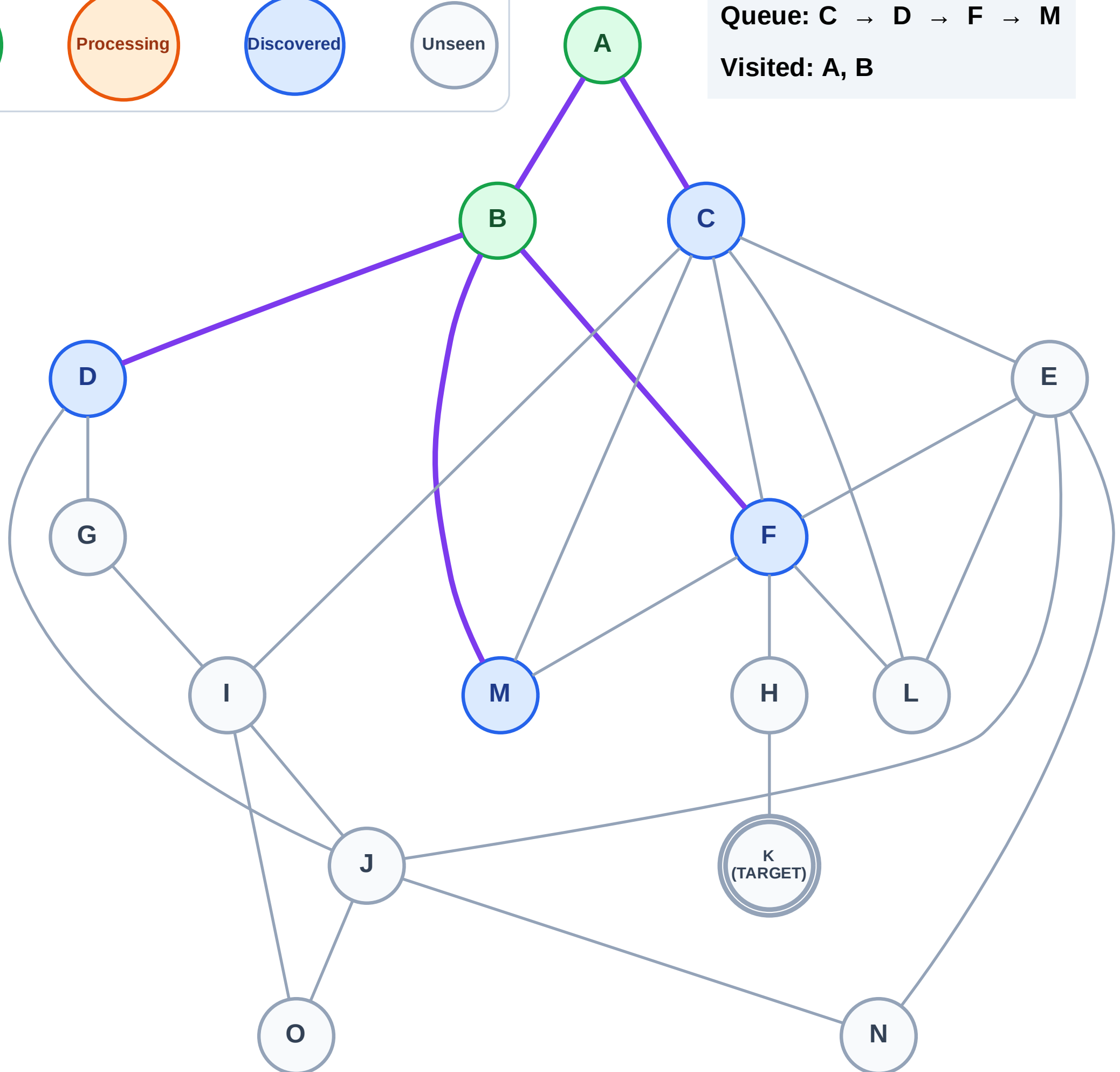
Step 5: Discover D, F, M from B

Legend



Queue: C → D → F → M

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

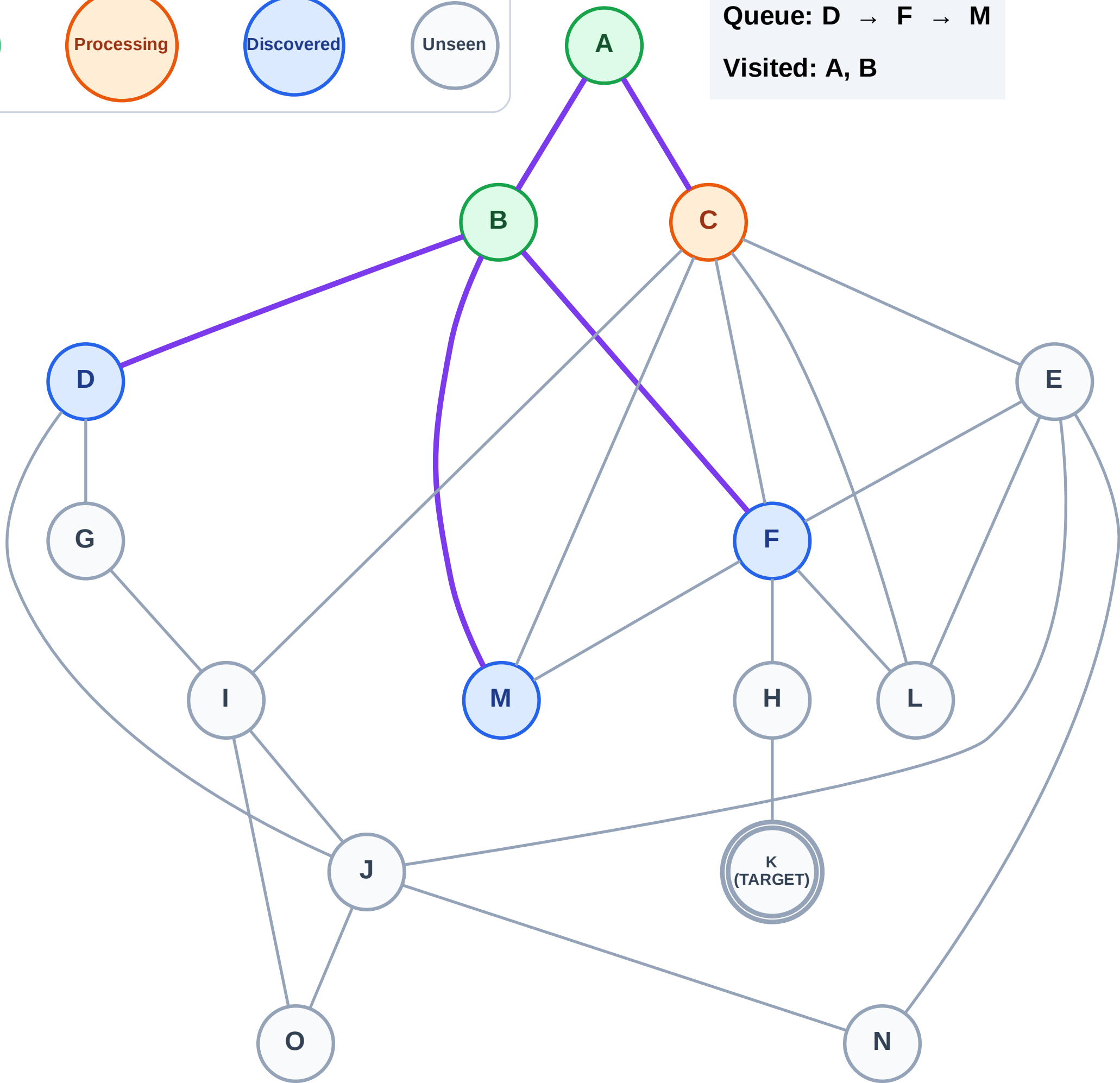
Processing

Discovered

Unseen

Queue: D → F → M

Visited: A, B



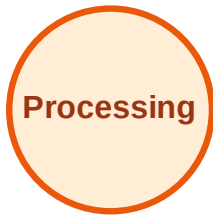
BFS Traversal

Step 7: Discover E, I, L from C

Legend



Complete



Processing



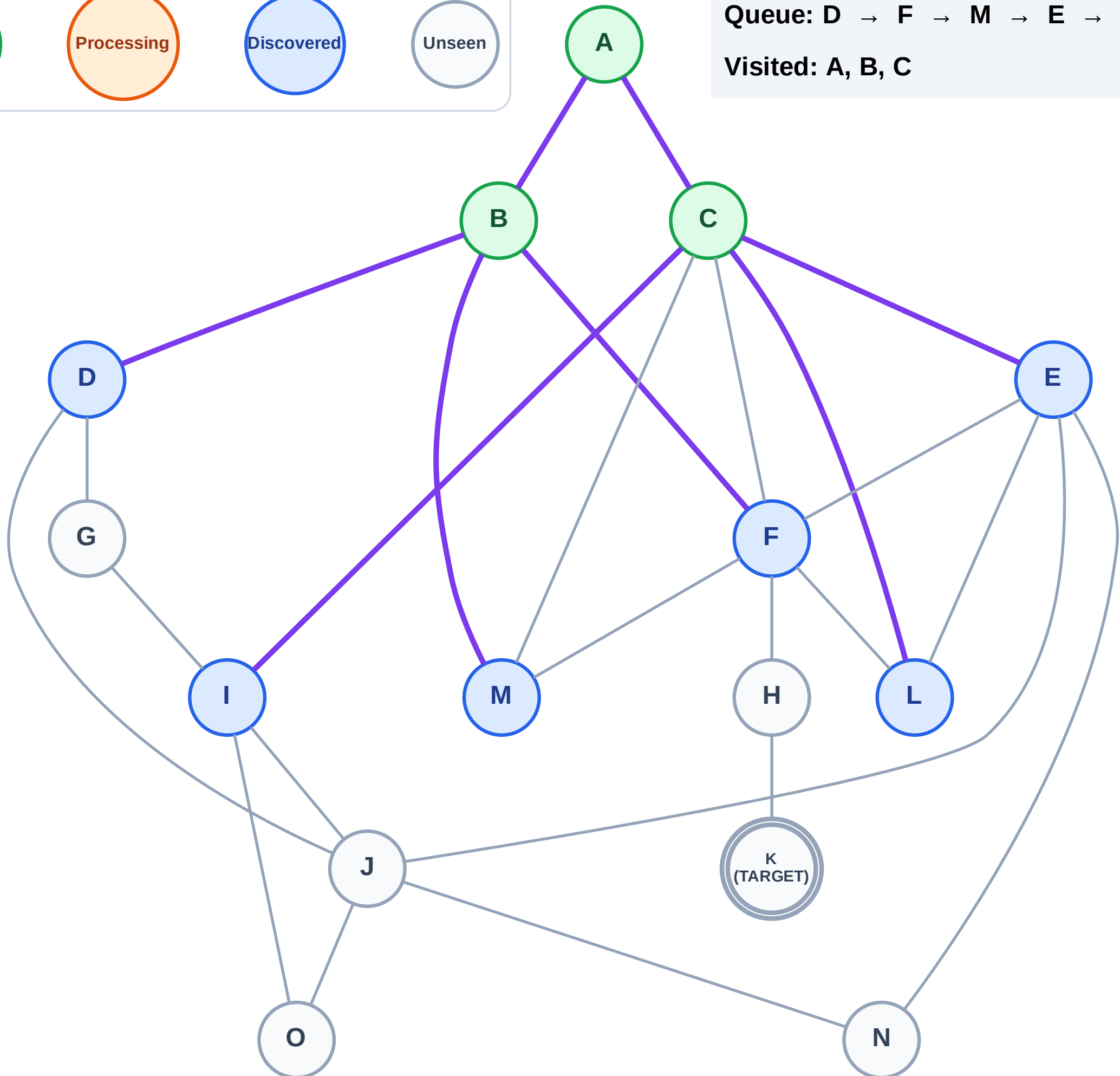
Discovered



Unseen

Queue: D → F → M → E → I → L

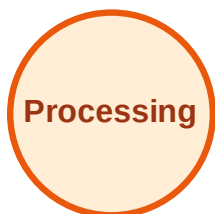
Visited: A, B, C



BFS Traversal

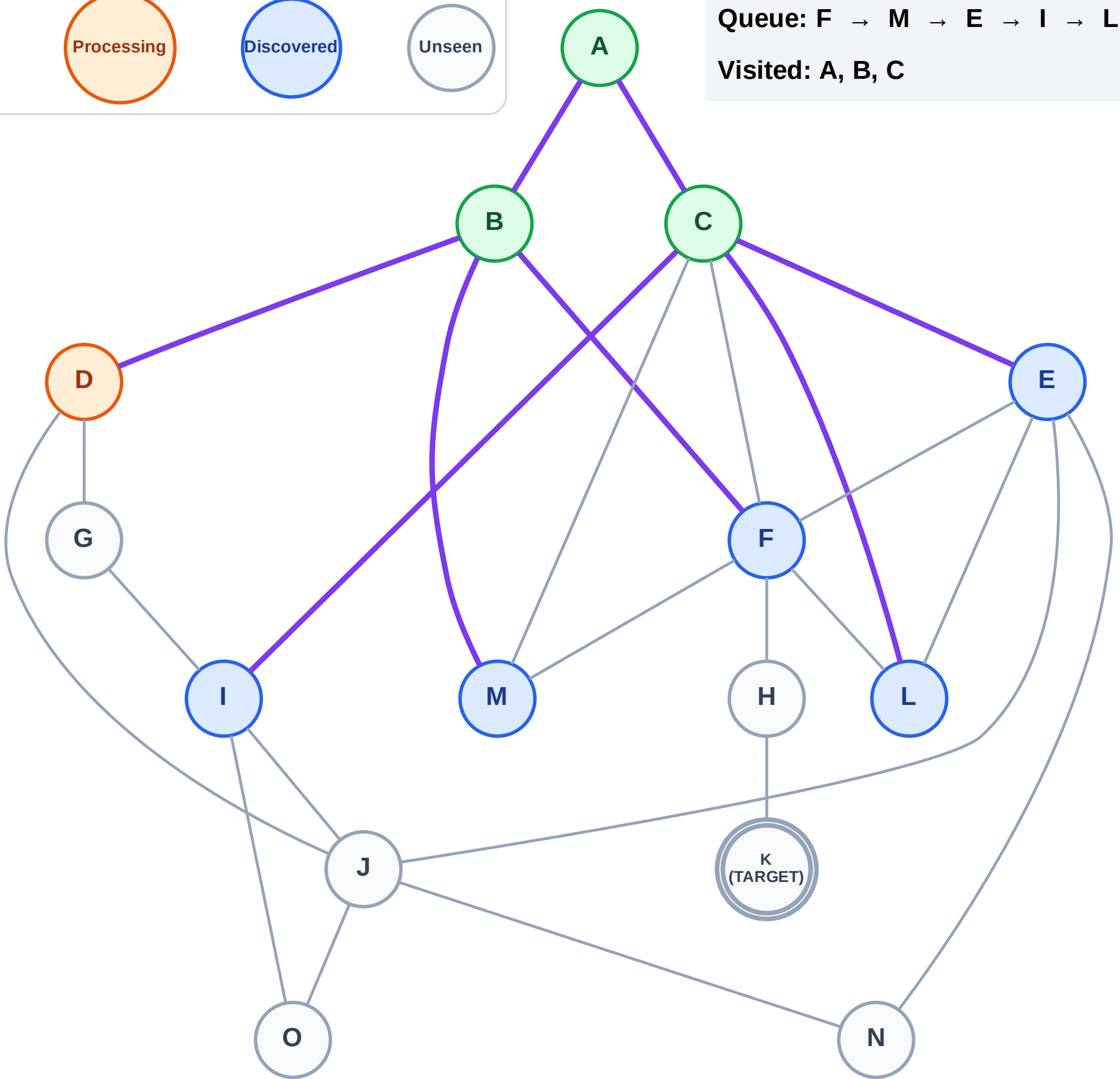
Step 8: Process node D

Legend



Queue: F → M → E → I → L

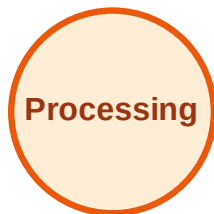
Visited: A, B, C



BFS Traversal

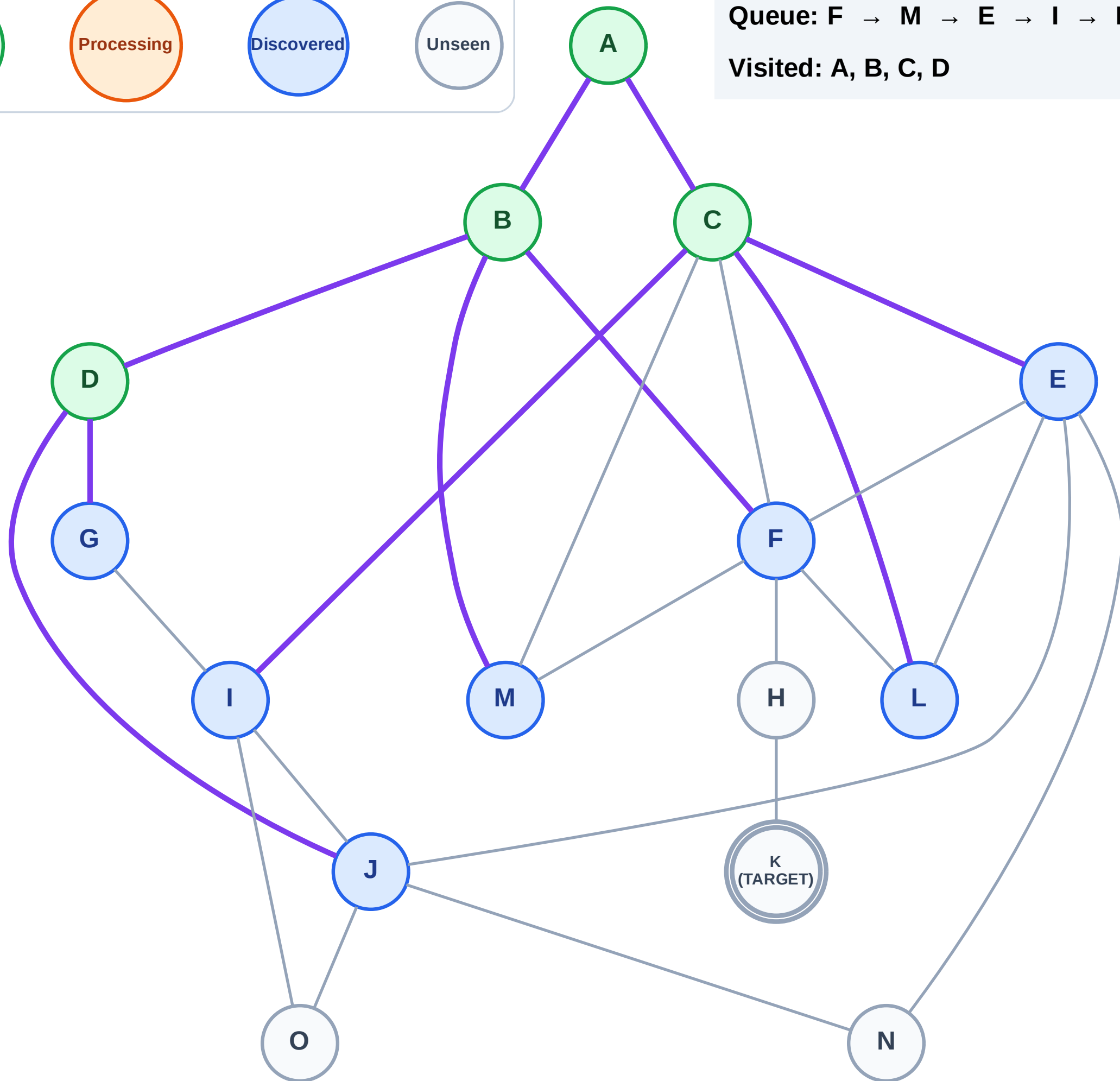
Step 9: Discover G, J from D

Legend



Queue: F → M → E → I → L → G → J

Visited: A, B, C, D



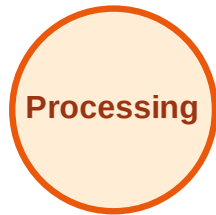
BFS Traversal

Step 10: Process node F

Legend



Complete



Processing



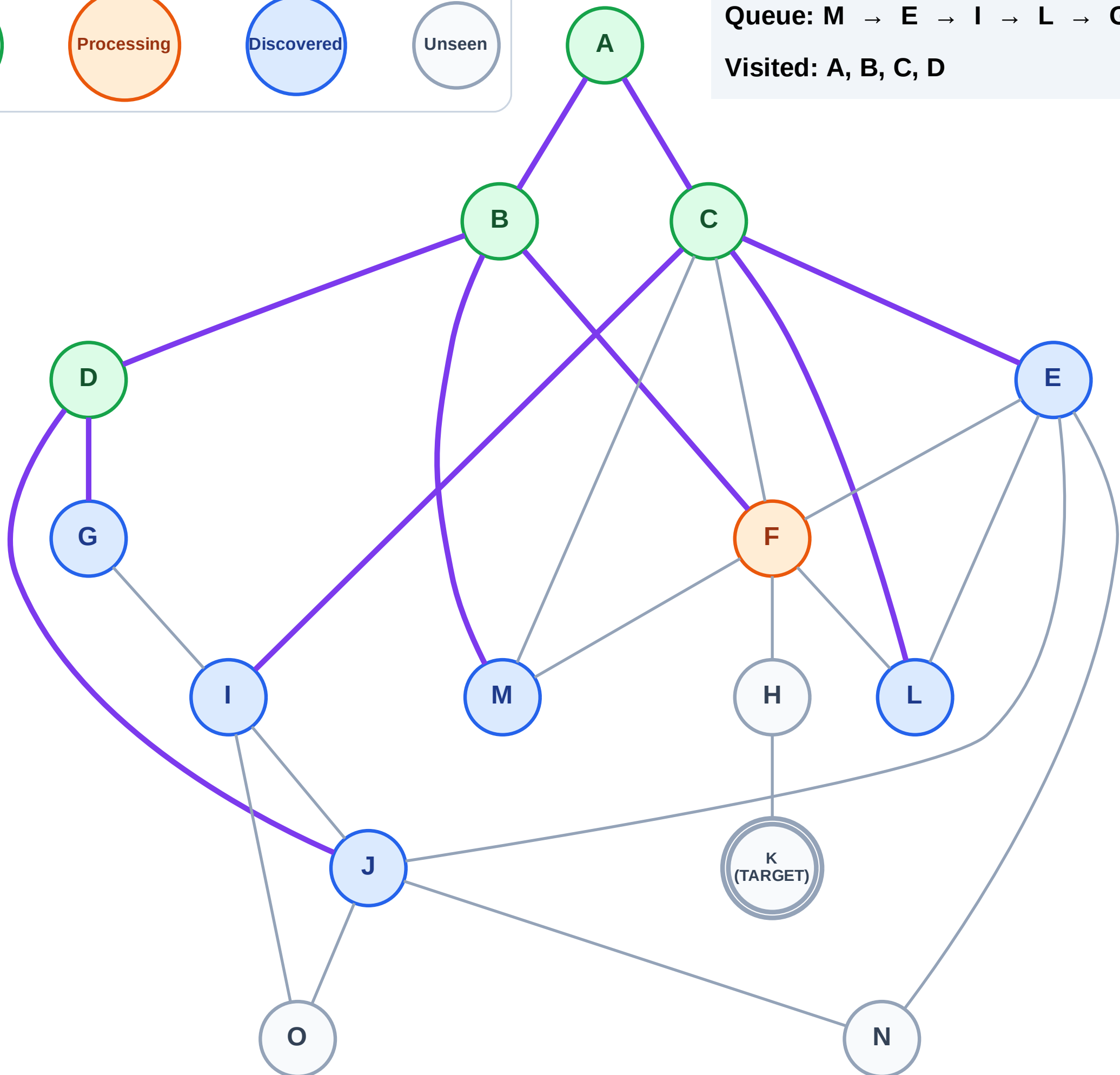
Discovered



Unseen

Queue: M → E → I → L → G → J

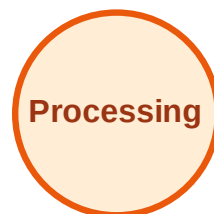
Visited: A, B, C, D



BFS Traversal

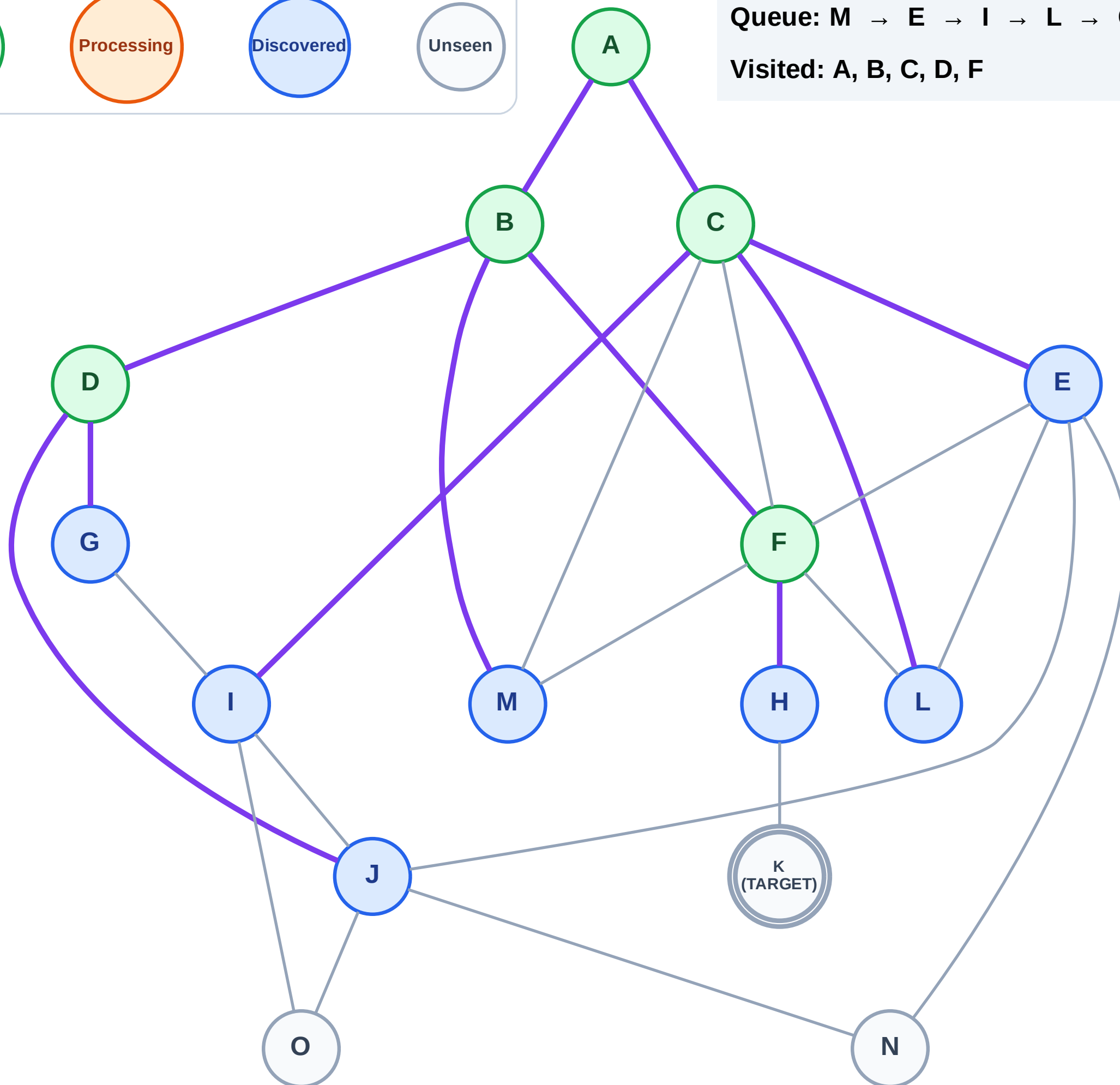
Step 11: Discover H from F

Legend



Queue: M → E → I → L → G → J → H

Visited: A, B, C, D, F



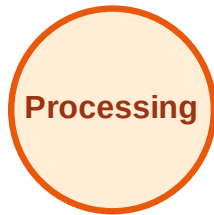
BFS Traversal

Step 12: Process node M

Legend



Complete



Processing



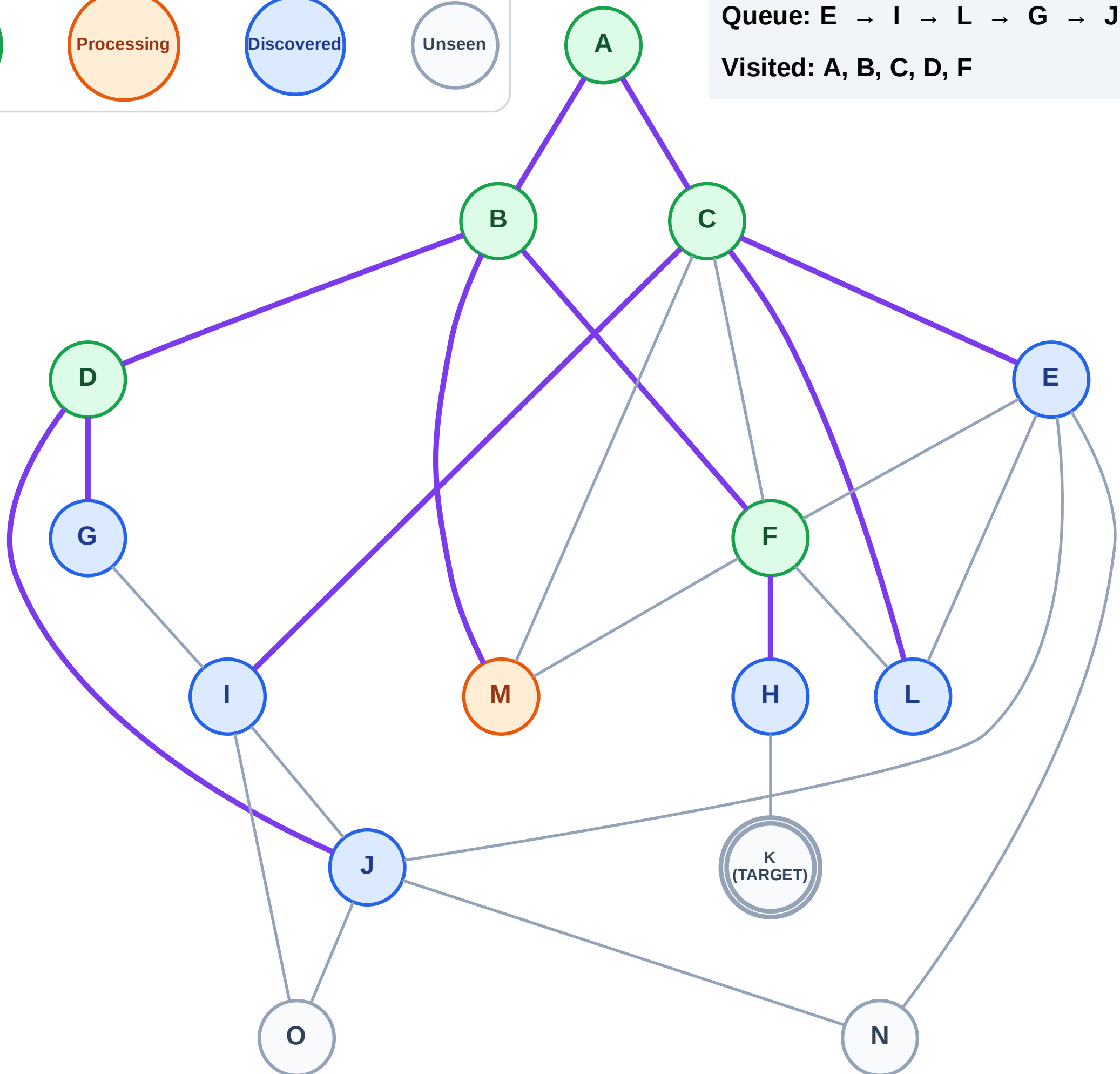
Discovered



Unseen

Queue: E → I → L → G → J → H

Visited: A, B, C, D, F



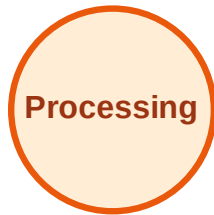
BFS Traversal

Step 13: No new neighbors from M

Legend



Complete



Processing



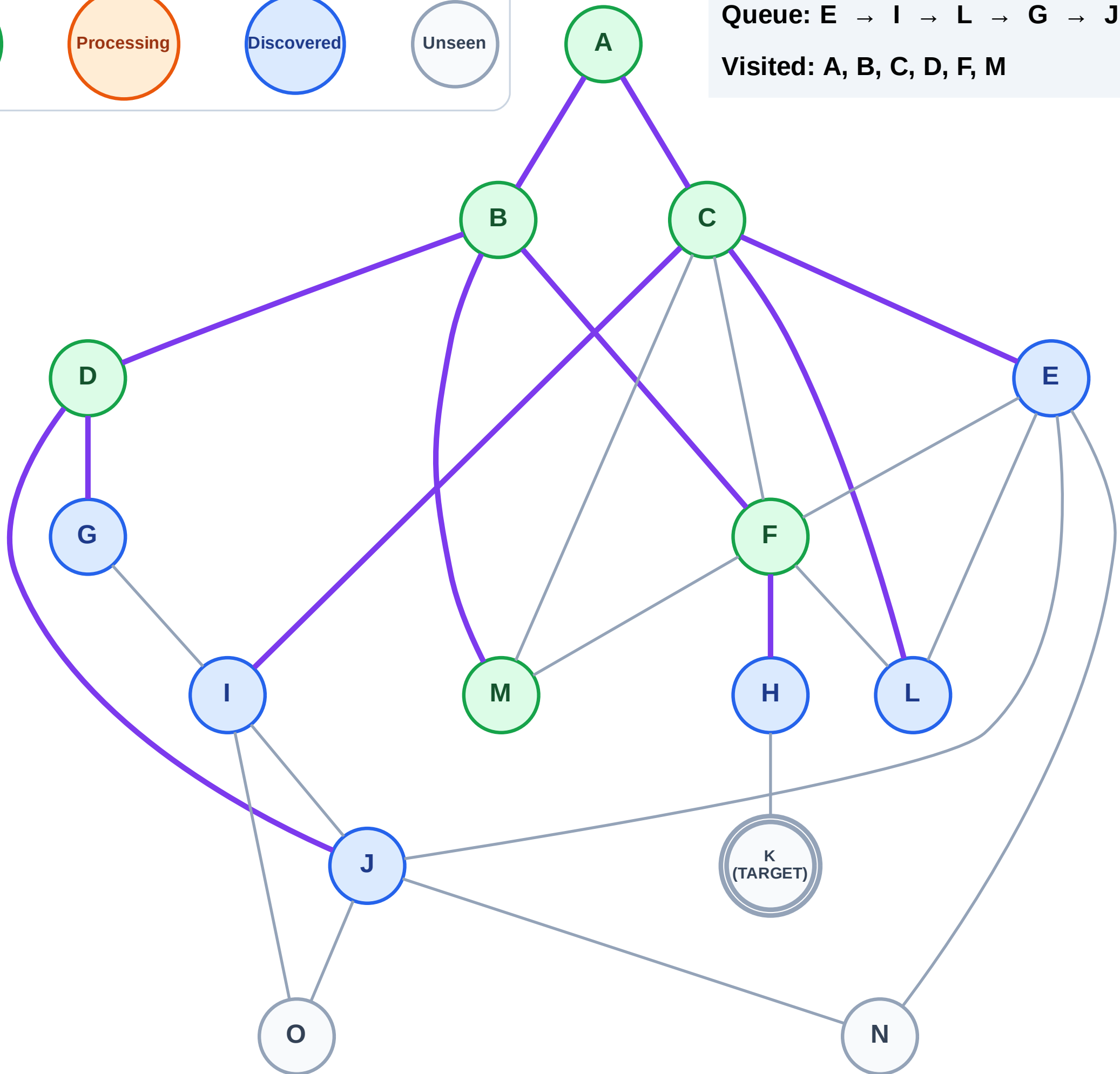
Discovered



Unseen

Queue: E → I → L → G → J → H

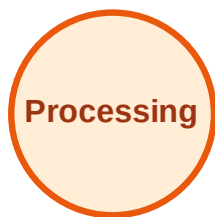
Visited: A, B, C, D, F, M



BFS Traversal

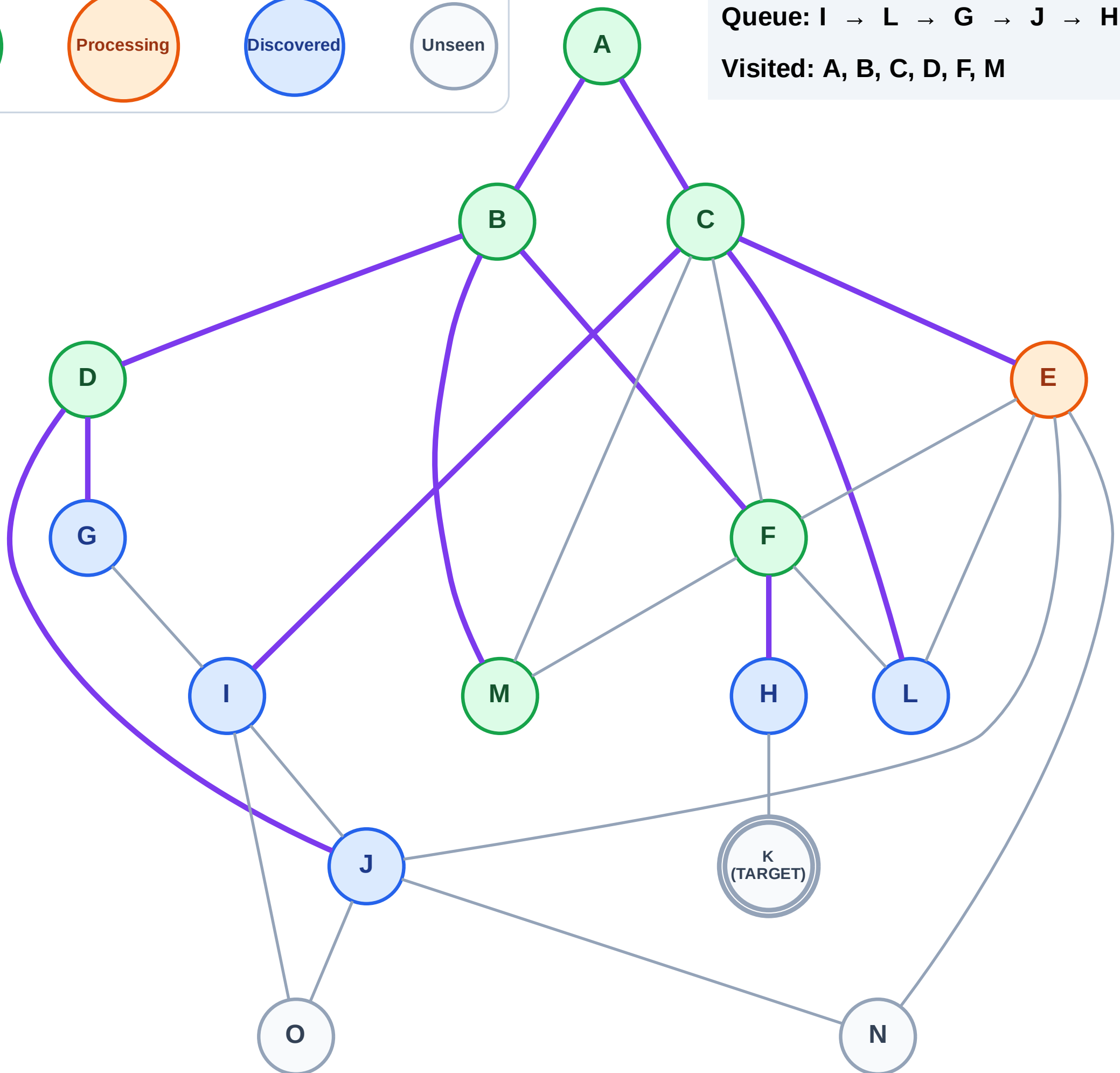
Step 14: Process node E

Legend



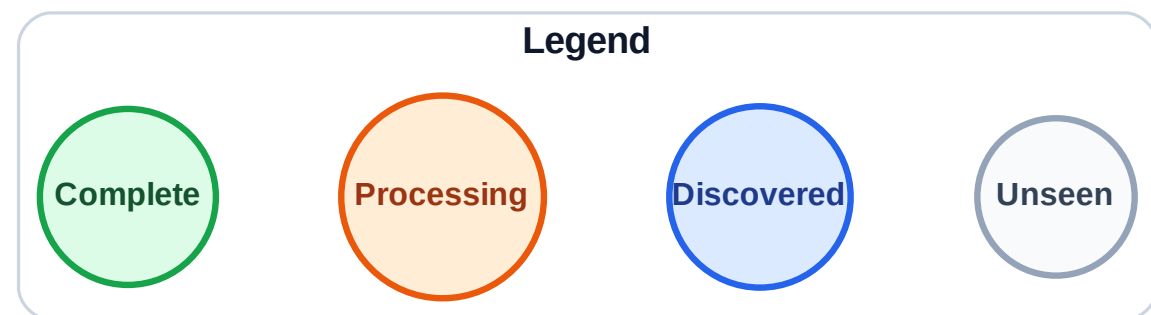
Queue: I → L → G → J → H

Visited: A, B, C, D, F, M



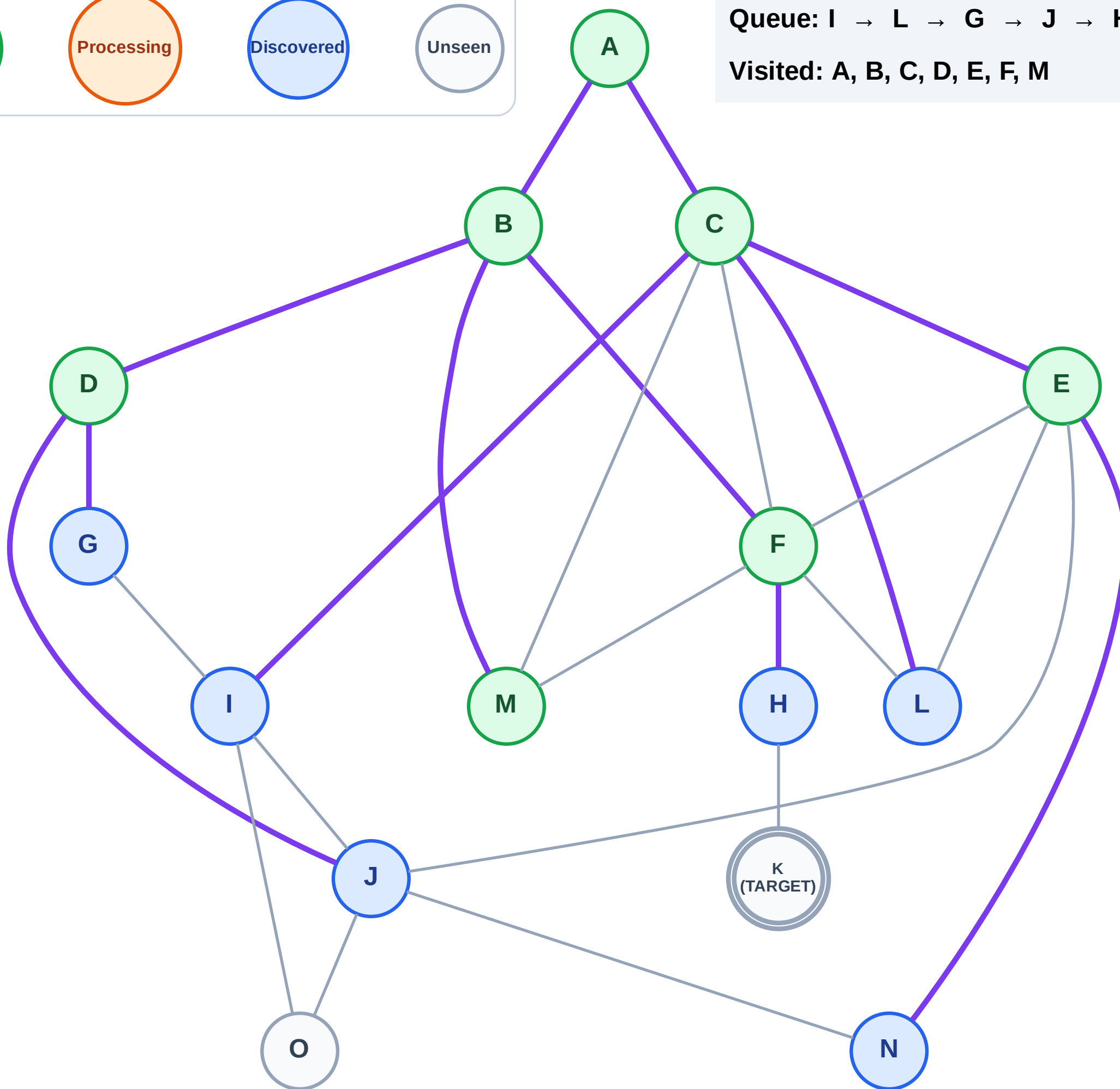
BFS Traversal

Step 15: Discover N from E



Queue: I → L → G → J → H → N

Visited: A, B, C, D, E, F, M



BFS Traversal

Step 16: Process node I

Legend

Complete

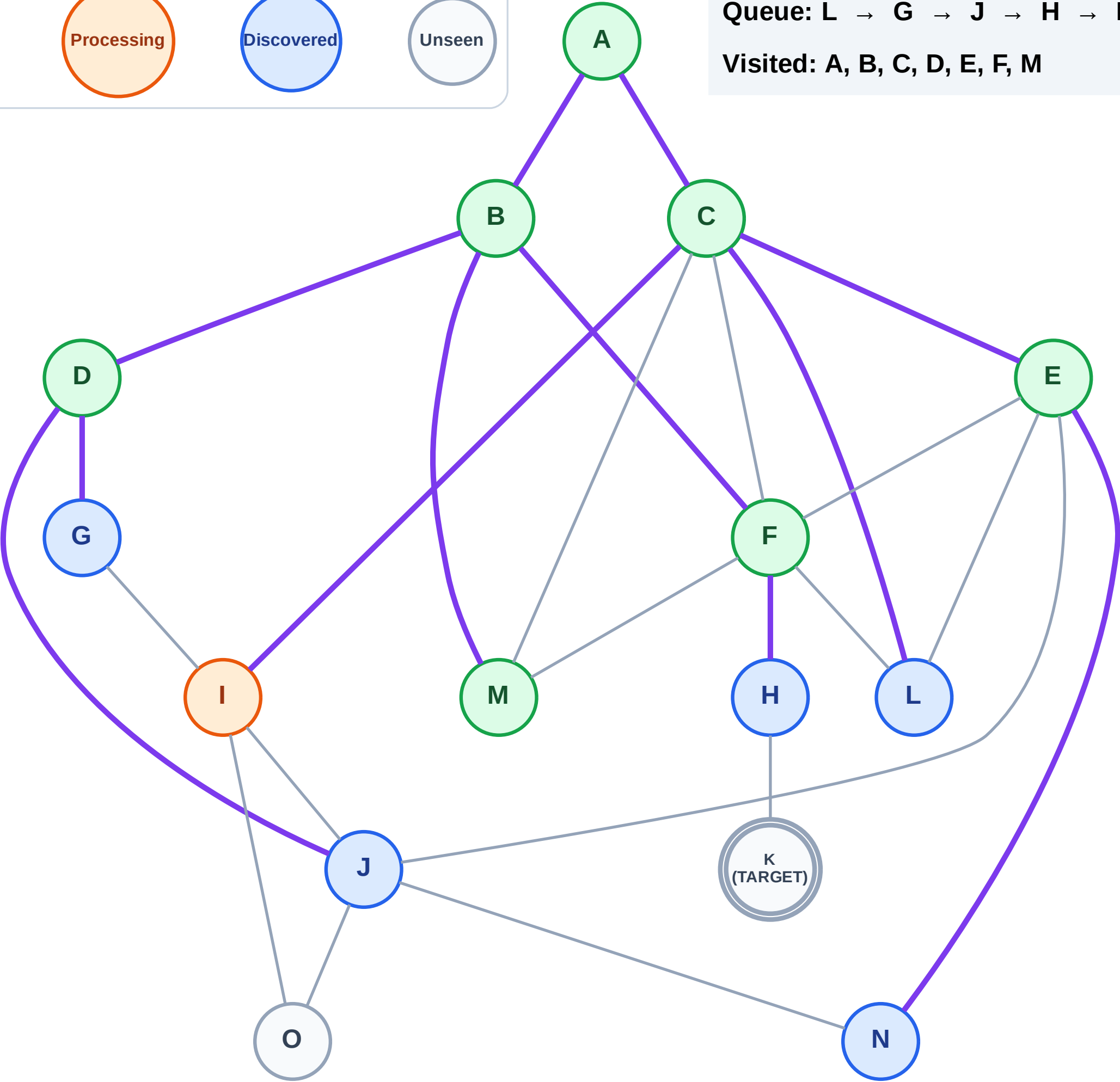
Processing

Discovered

Unseen

Queue: L → G → J → H → N

Visited: A, B, C, D, E, F, M



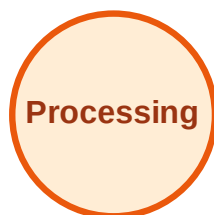
BFS Traversal

Step 17: Discover O from I

Legend



Complete



Processing



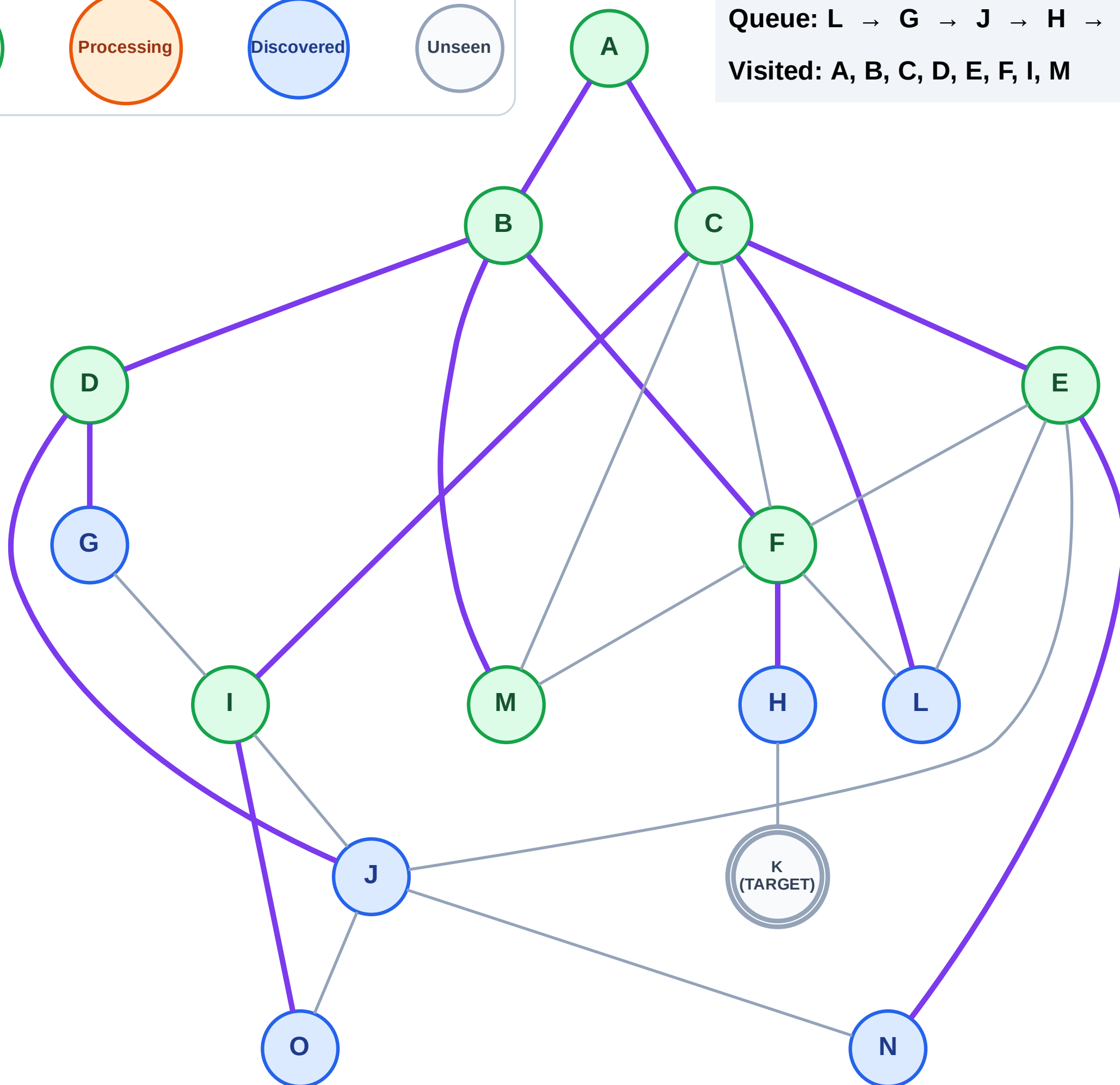
Discovered



Unseen

Queue: L → G → J → H → N → O

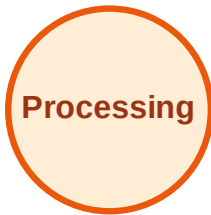
Visited: A, B, C, D, E, F, I, M



BFS Traversal

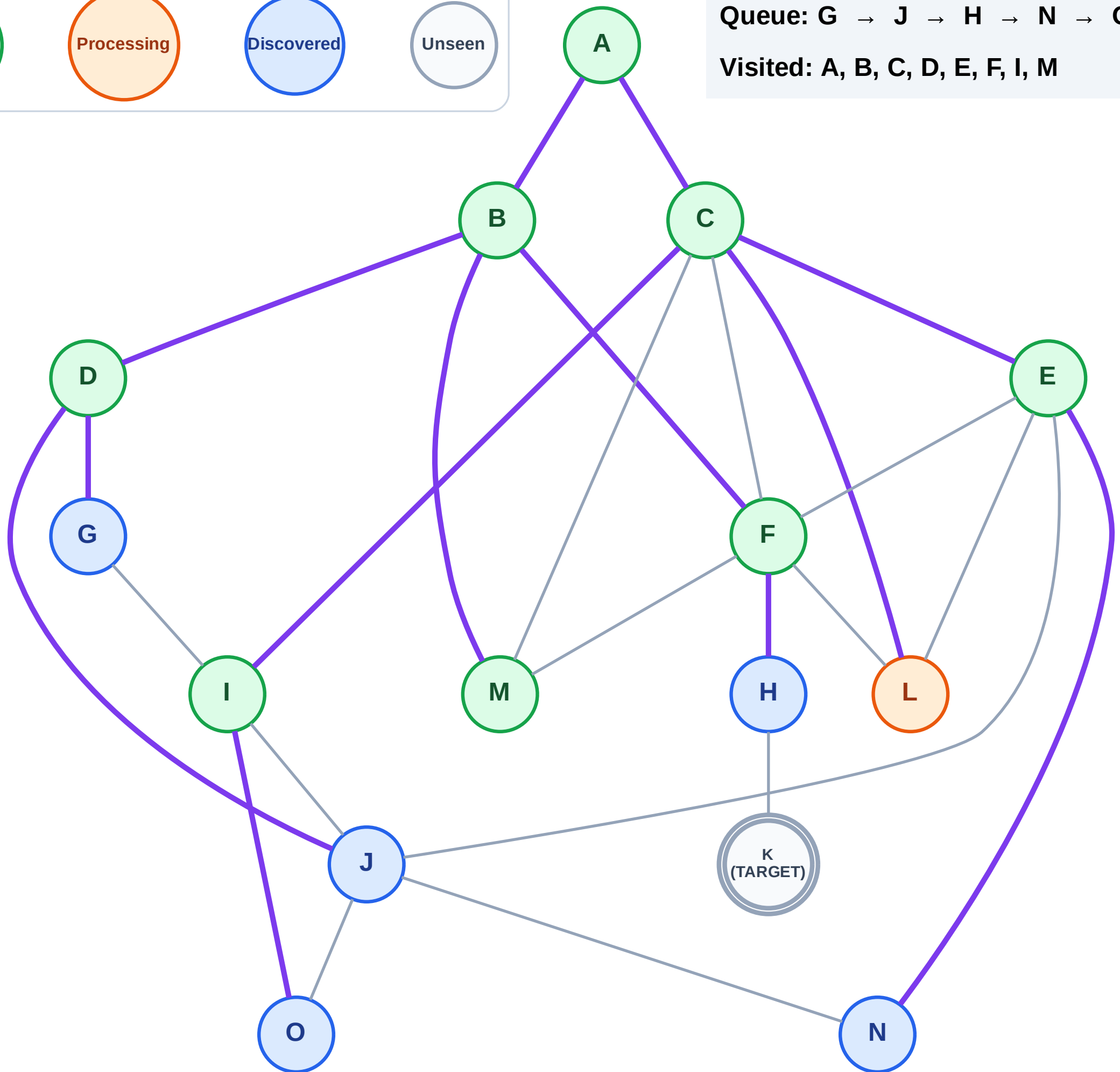
Step 18: Process node L

Legend



Queue: G → J → H → N → O

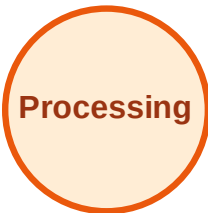
Visited: A, B, C, D, E, F, I, M



BFS Traversal

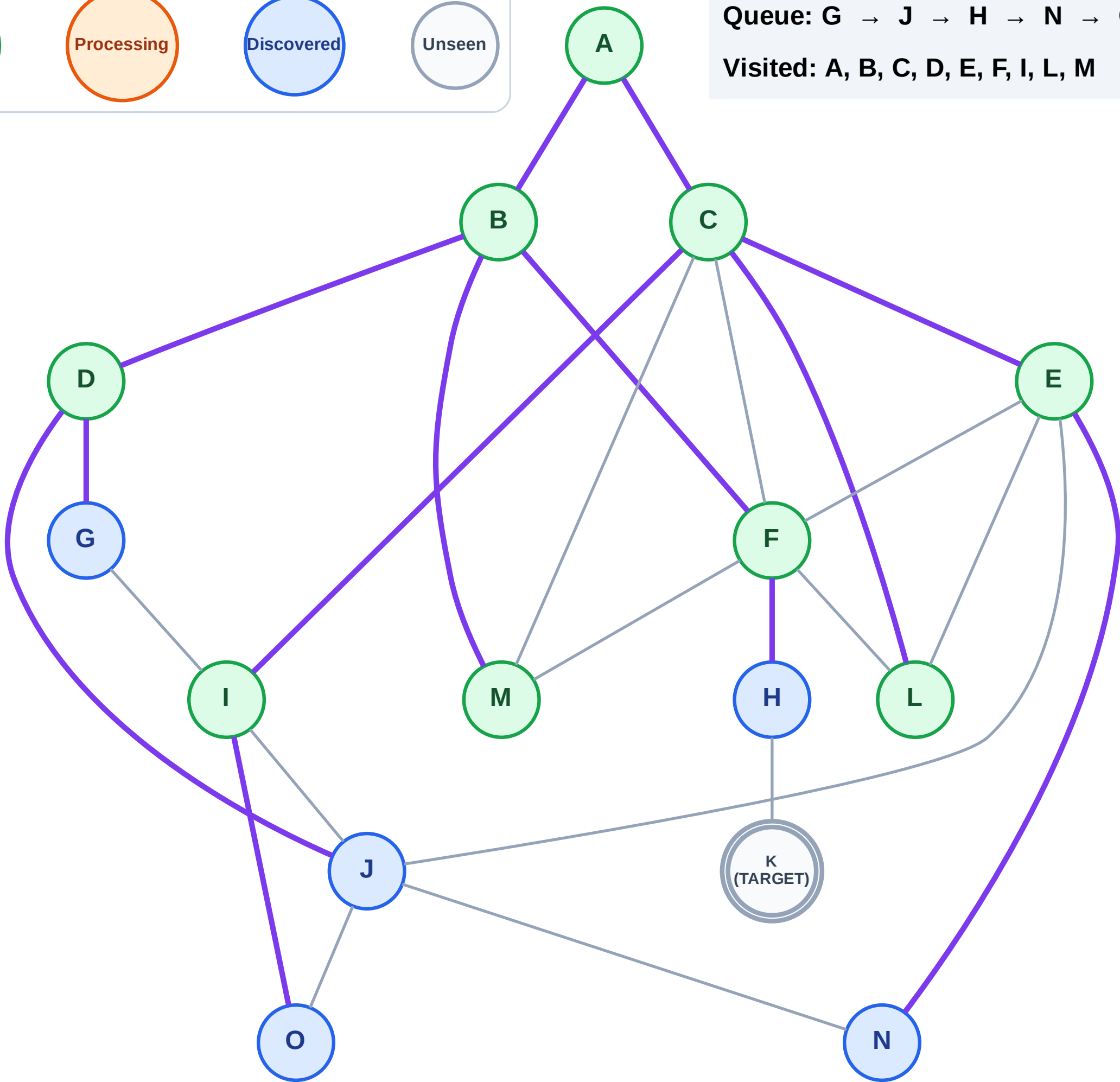
Step 19: No new neighbors from L

Legend



Queue: G → J → H → N → O

Visited: A, B, C, D, E, F, I, L, M



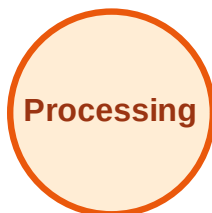
BFS Traversal

Step 20: Process node G

Legend



Complete



Processing



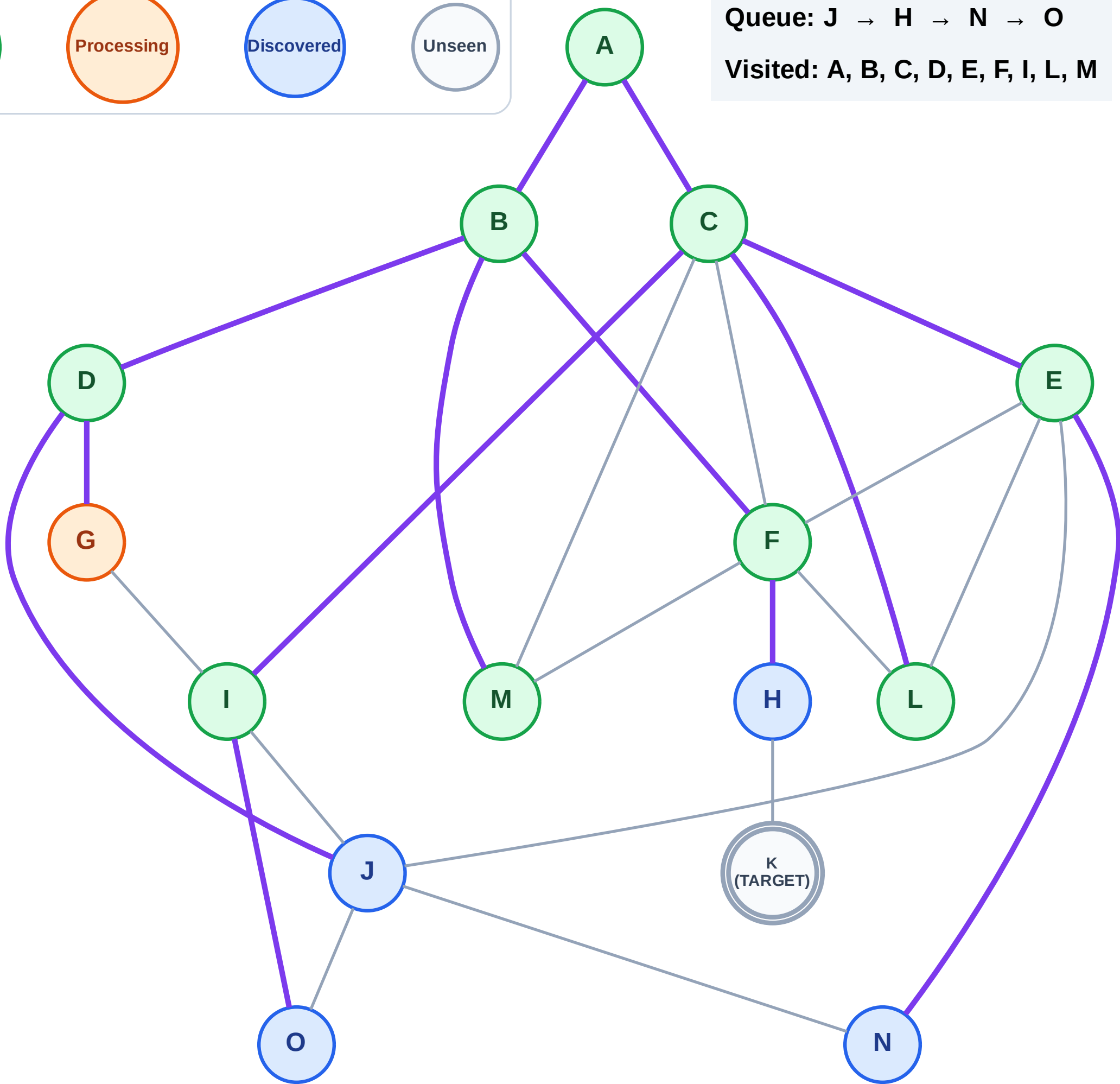
Discovered



Unseen

Queue: J → H → N → O

Visited: A, B, C, D, E, F, I, L, M



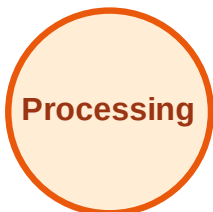
BFS Traversal

Step 21: No new neighbors from G

Legend



Complete



Processing



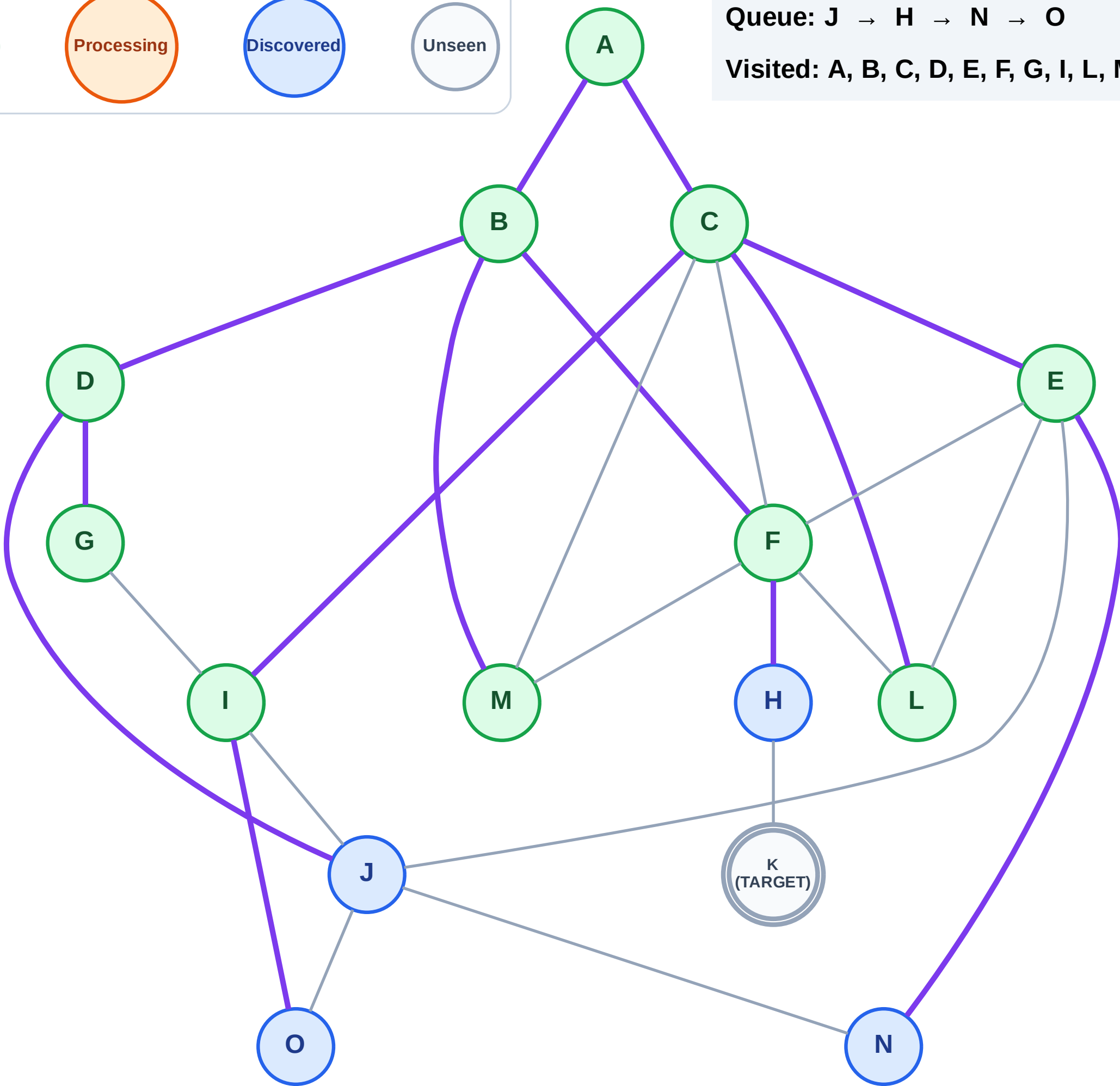
Discovered



Unseen

Queue: J → H → N → O

Visited: A, B, C, D, E, F, G, I, L, M



BFS Traversal

Step 22: Process node J

Legend



Complete



Processing



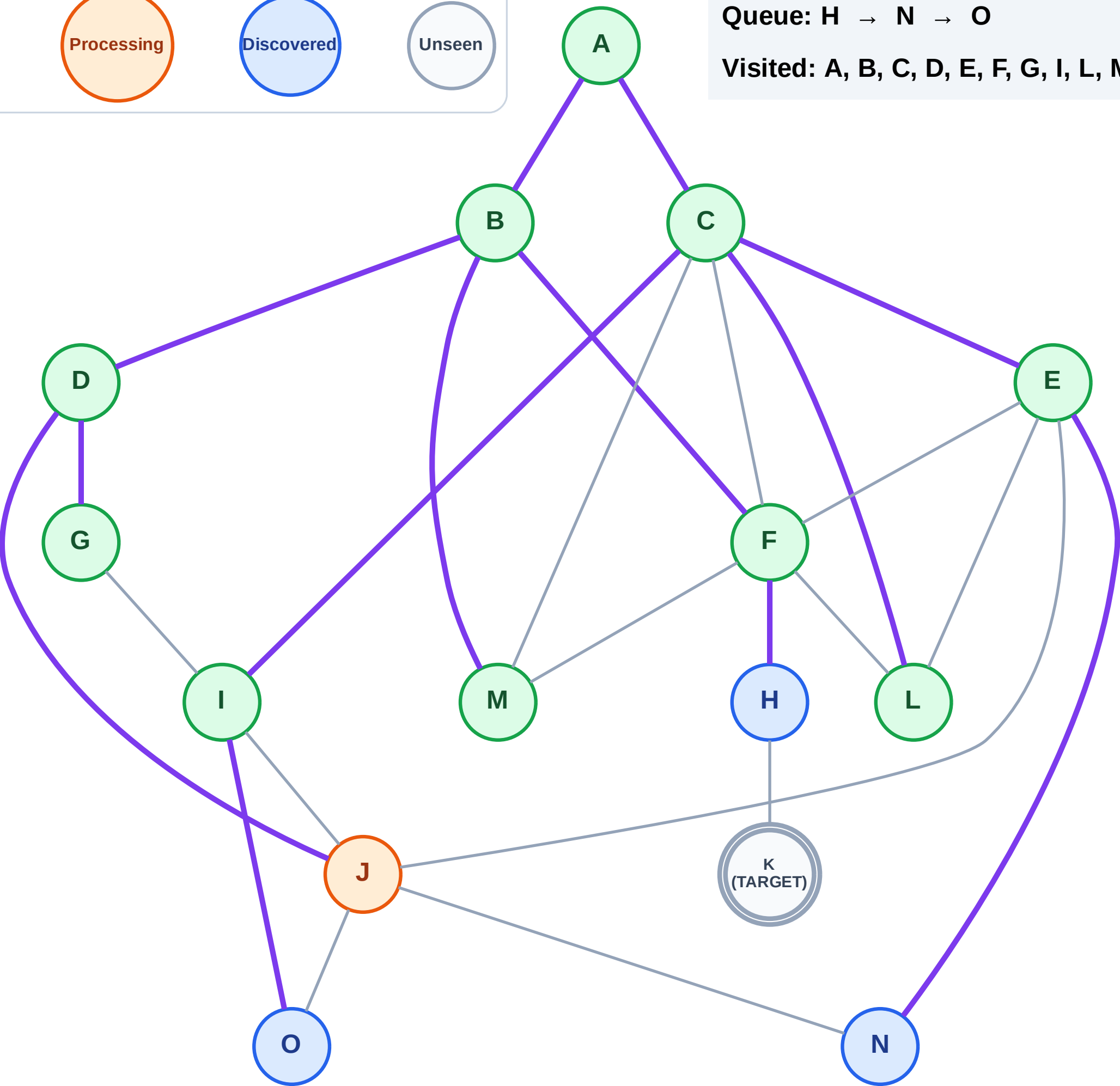
Discovered



Unseen

Queue: H → N → O

Visited: A, B, C, D, E, F, G, I, L, M



BFS Traversal

Step 23: No new neighbors from J

Legend

Complete

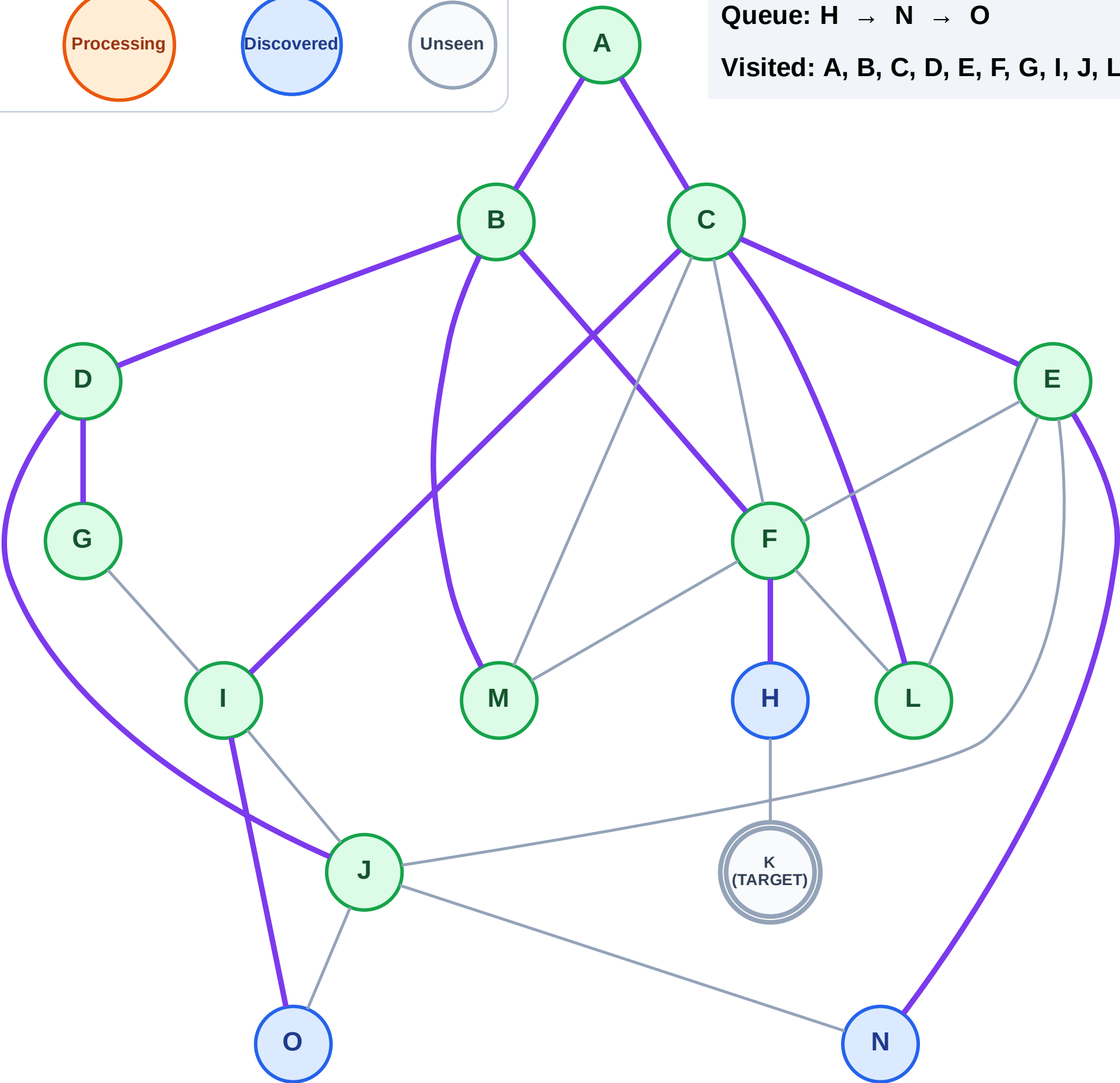
Processing

Discovered

Unseen

Queue: H → N → O

Visited: A, B, C, D, E, F, G, I, J, L, M



BFS Traversal

Step 24: Process node H

Legend

Complete

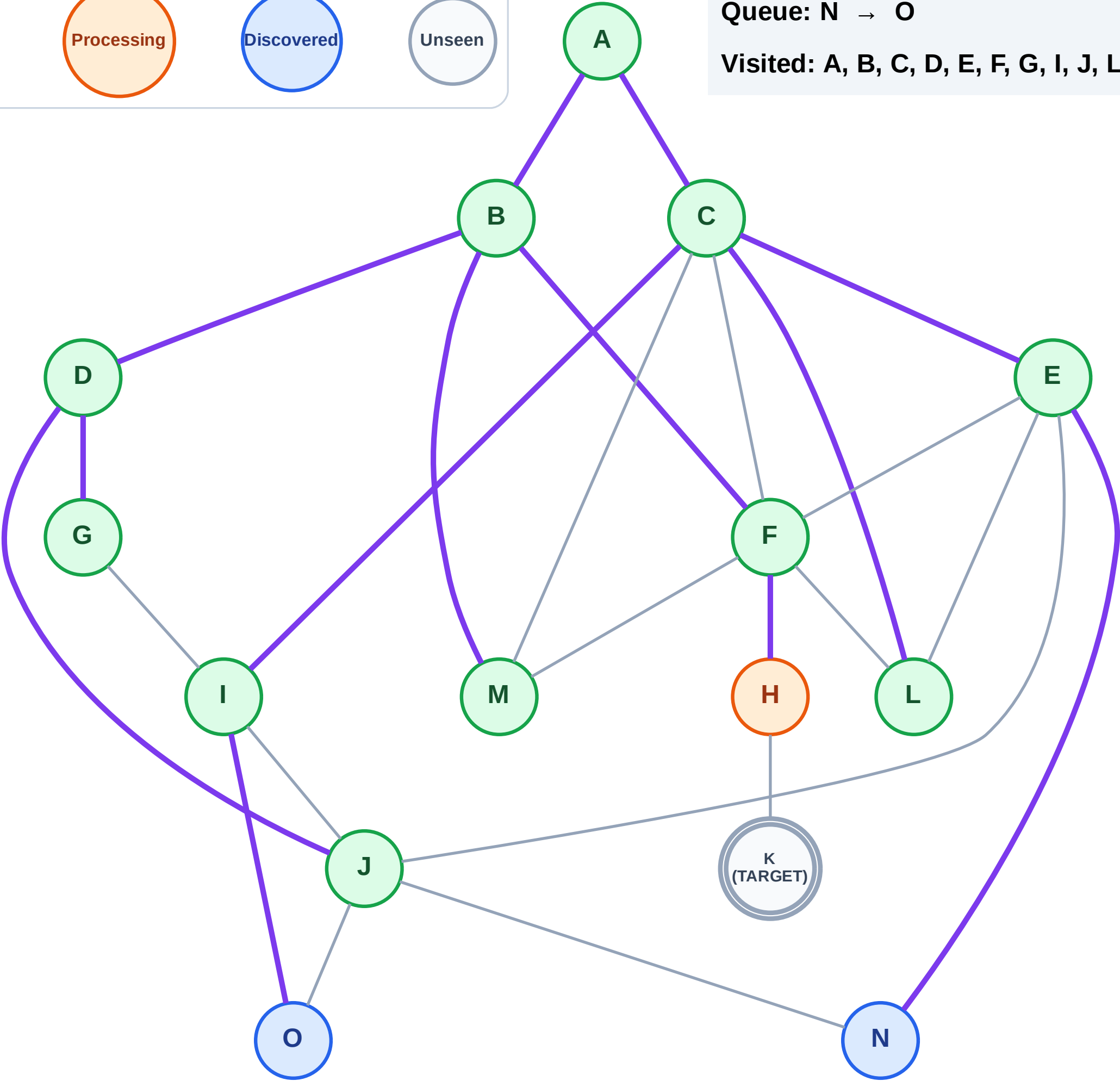
Processing

Discovered

Unseen

Queue: N → O

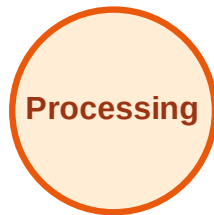
Visited: A, B, C, D, E, F, G, I, J, L, M



BFS Traversal

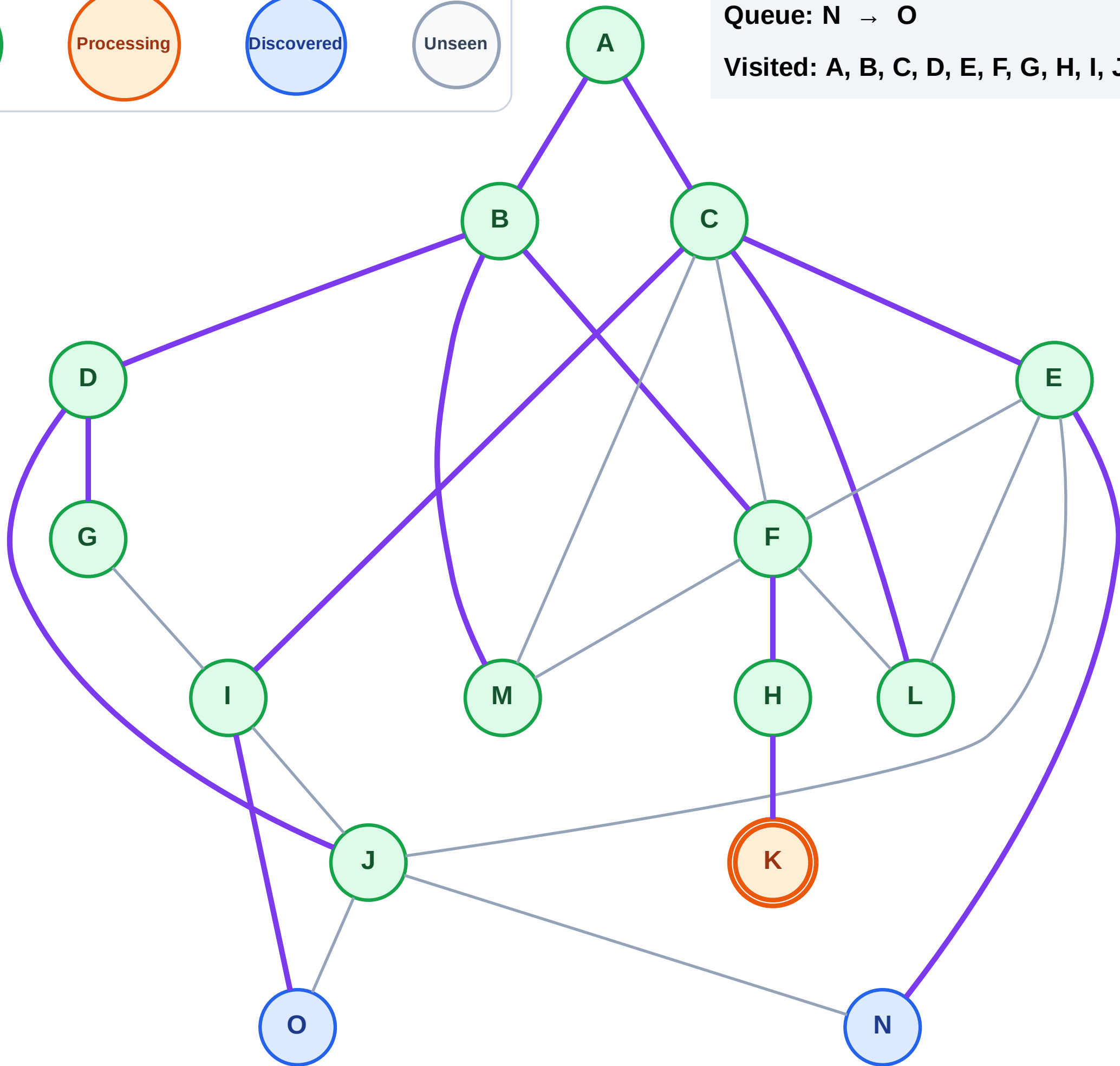
Step 25: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → F → H → K

Legend



Queue: N → O

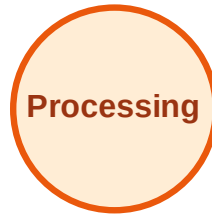
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M



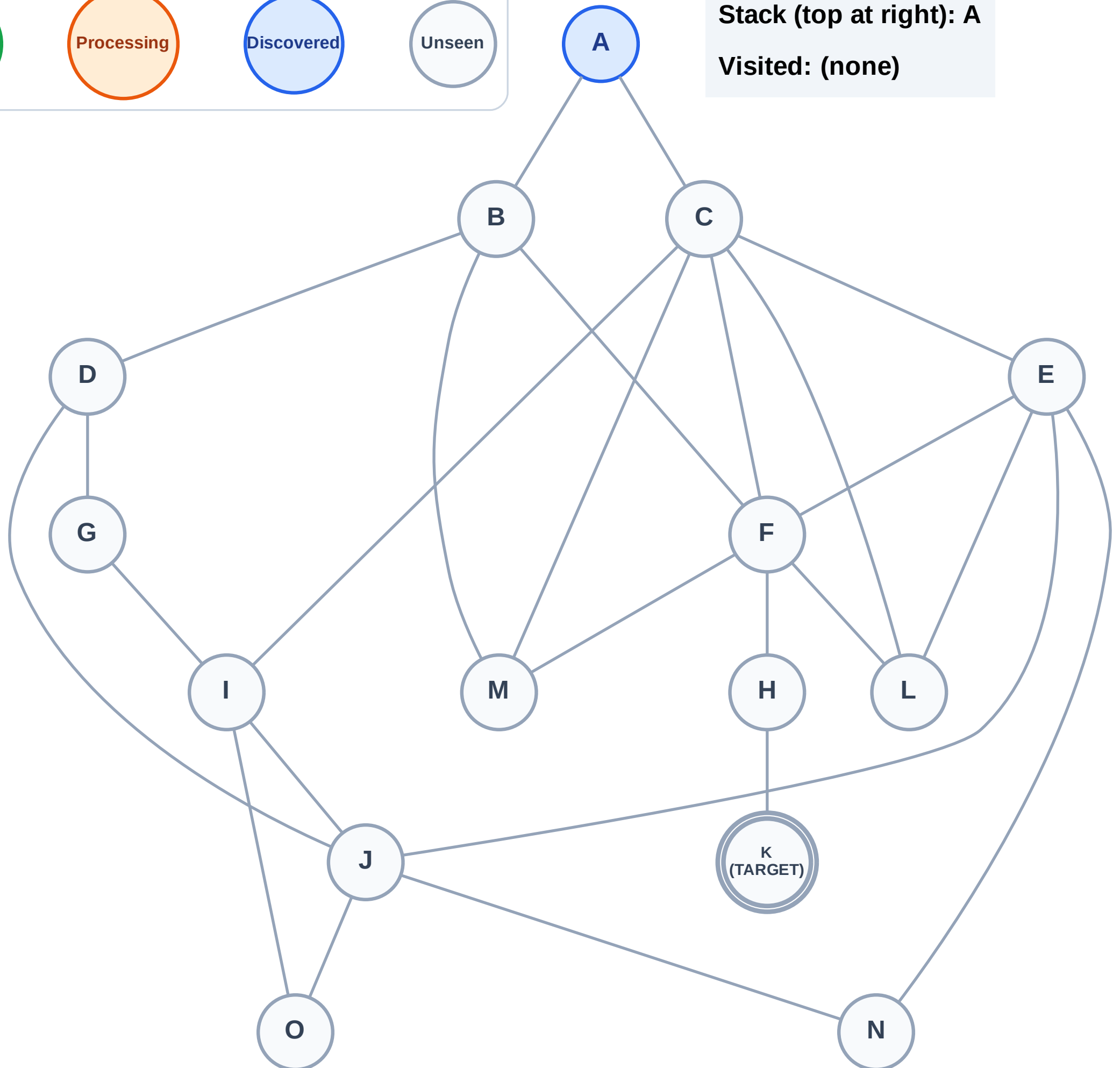
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

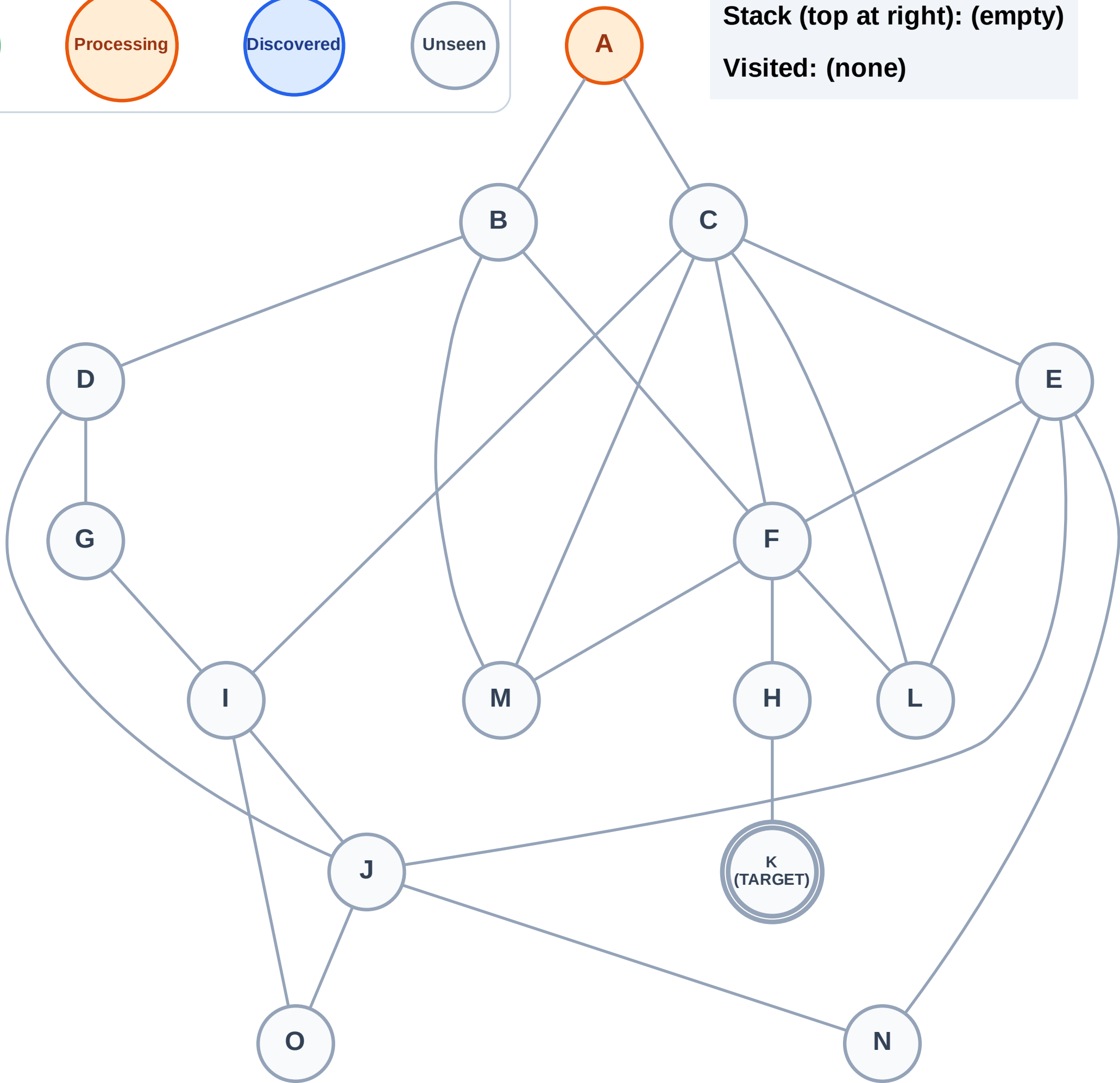
Complete

Processing

Discovered

Unseen

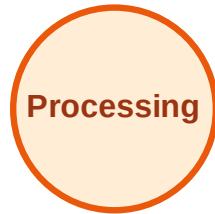
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

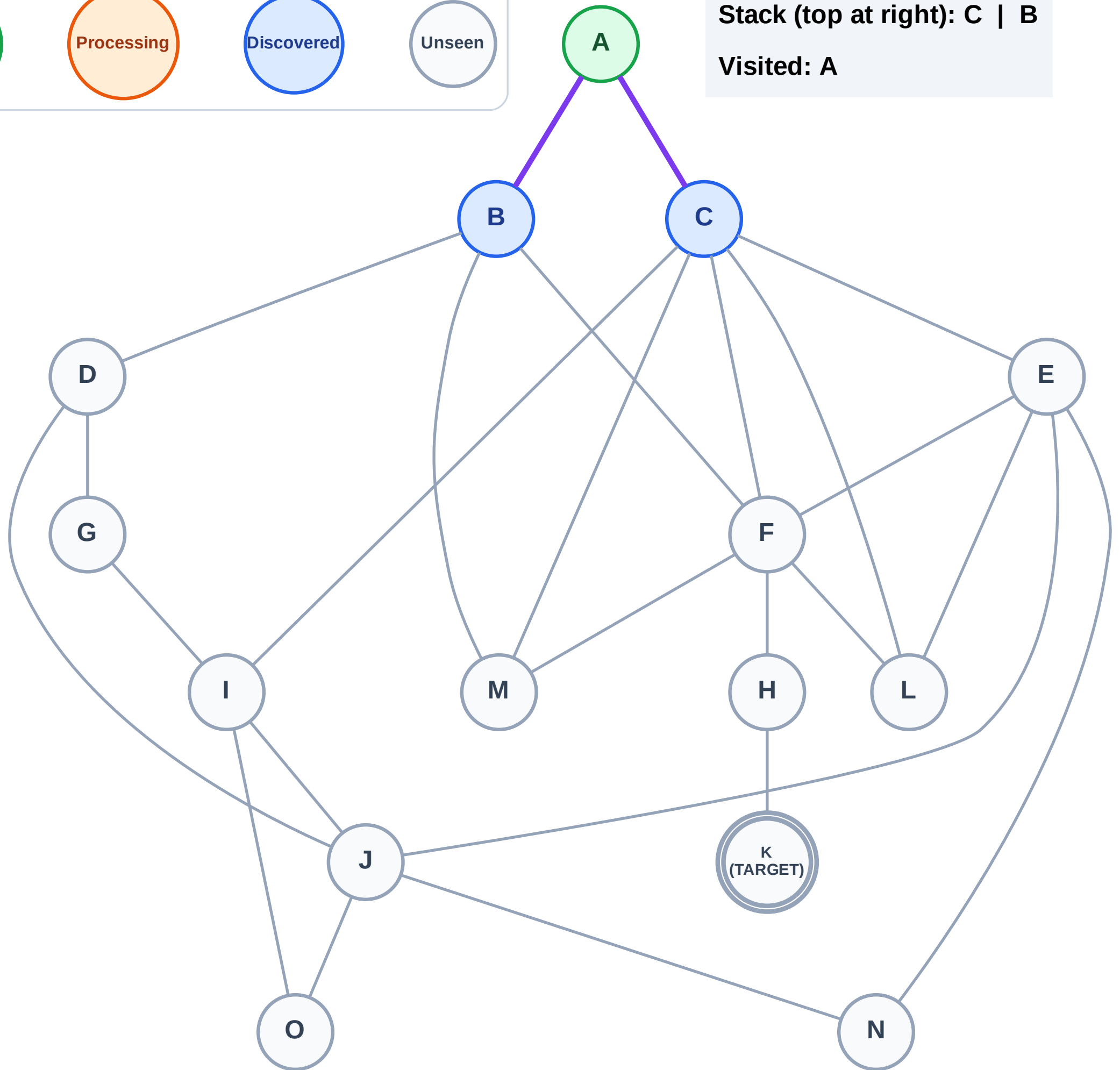
Step 3: Discover C, B from A

Legend



Stack (top at right): C | B

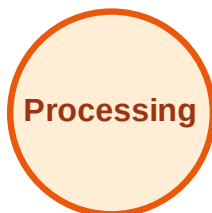
Visited: A



DFS Traversal

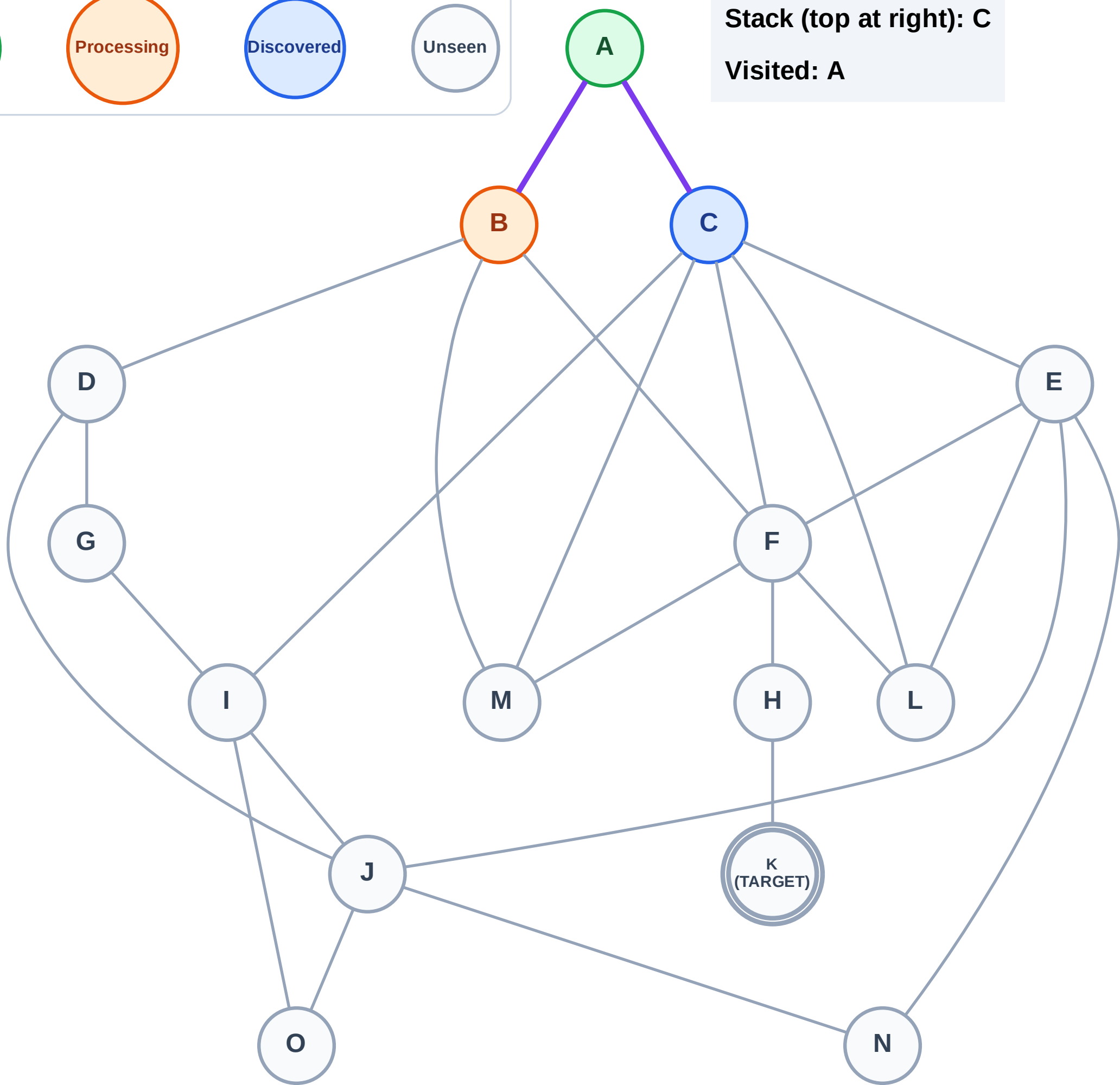
Step 4: Process node B

Legend



Stack (top at right): C

Visited: A

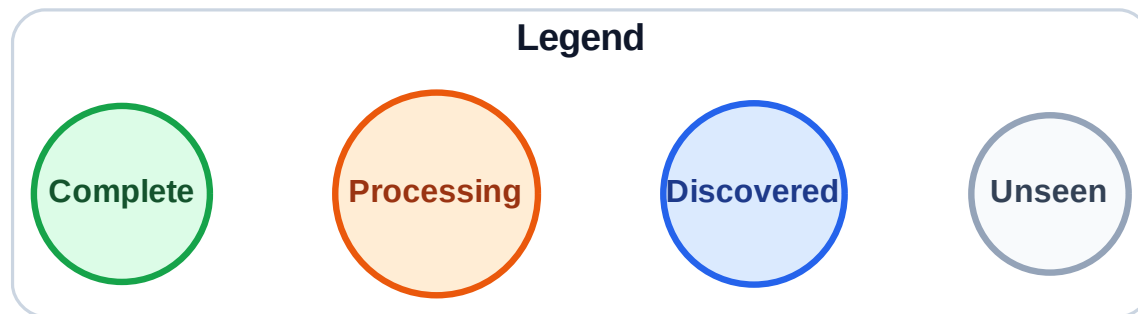


DFS Traversal

Step 5: Discover M, F, D from B

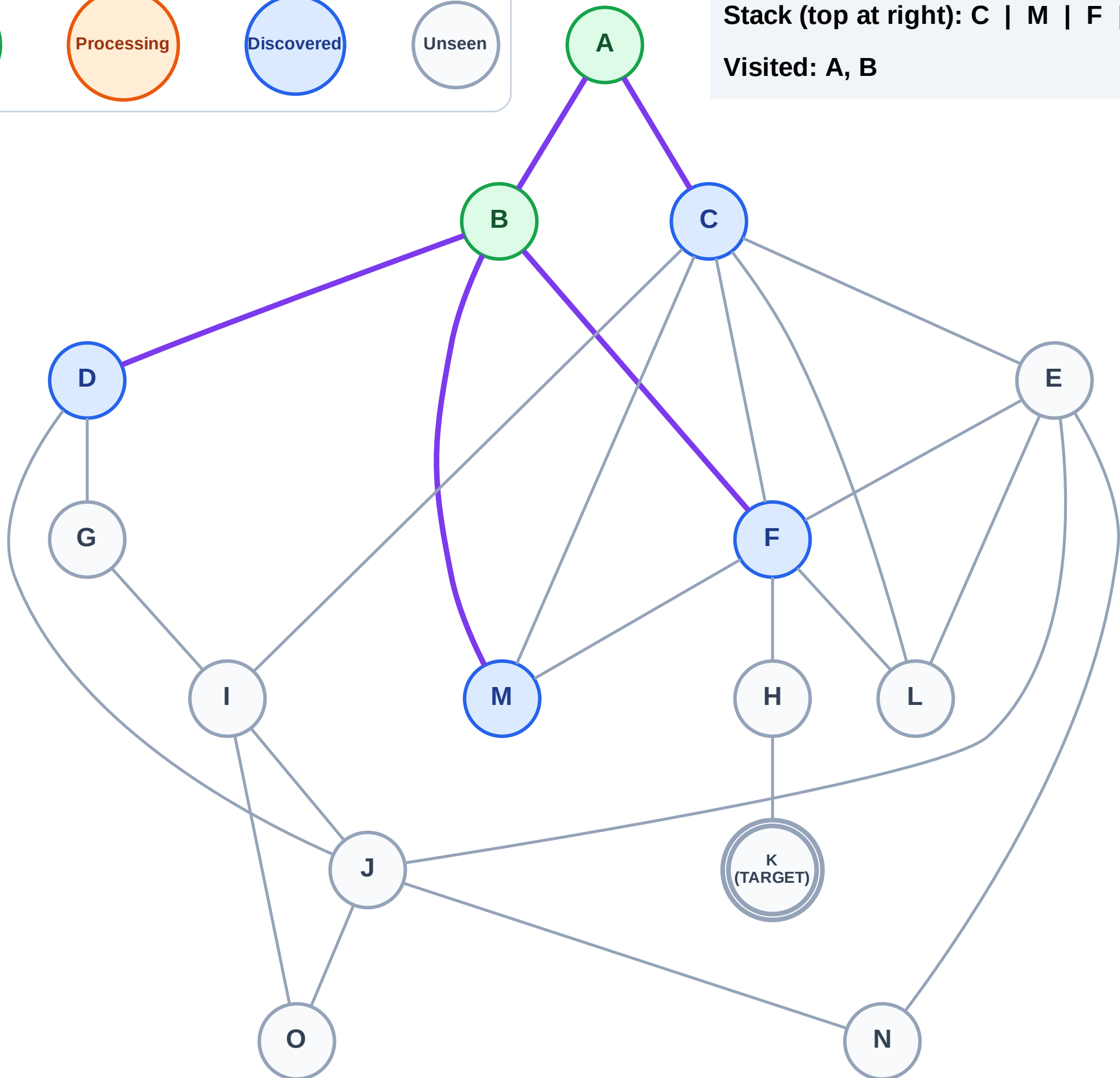
DFS Traversal

Step 5: Discover M, F, D from B



Stack (top at right): C | M | F | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

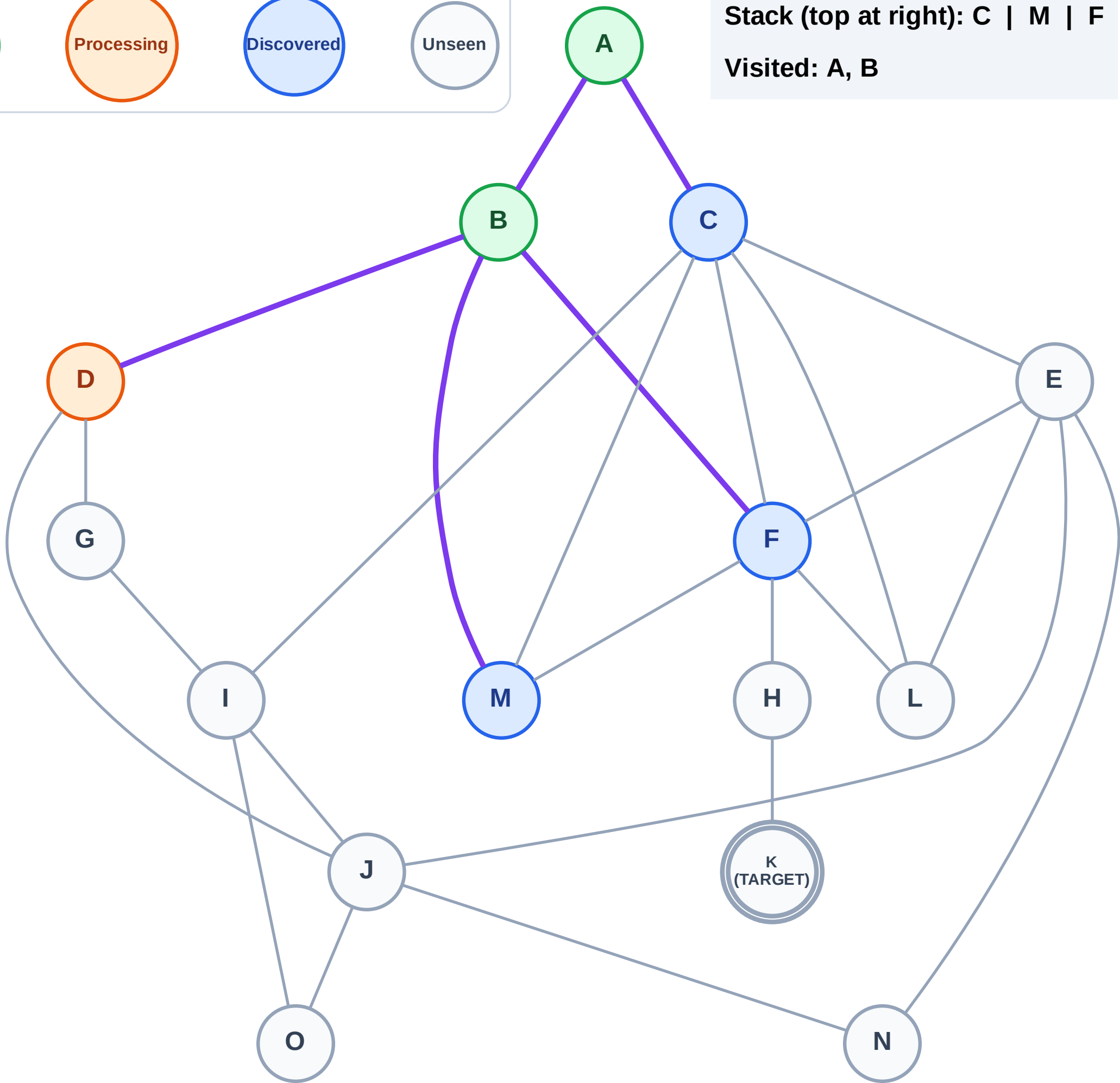
Processing

Discovered

Unseen

Stack (top at right): C | M | F

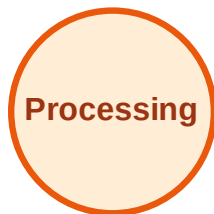
Visited: A, B



DFS Traversal

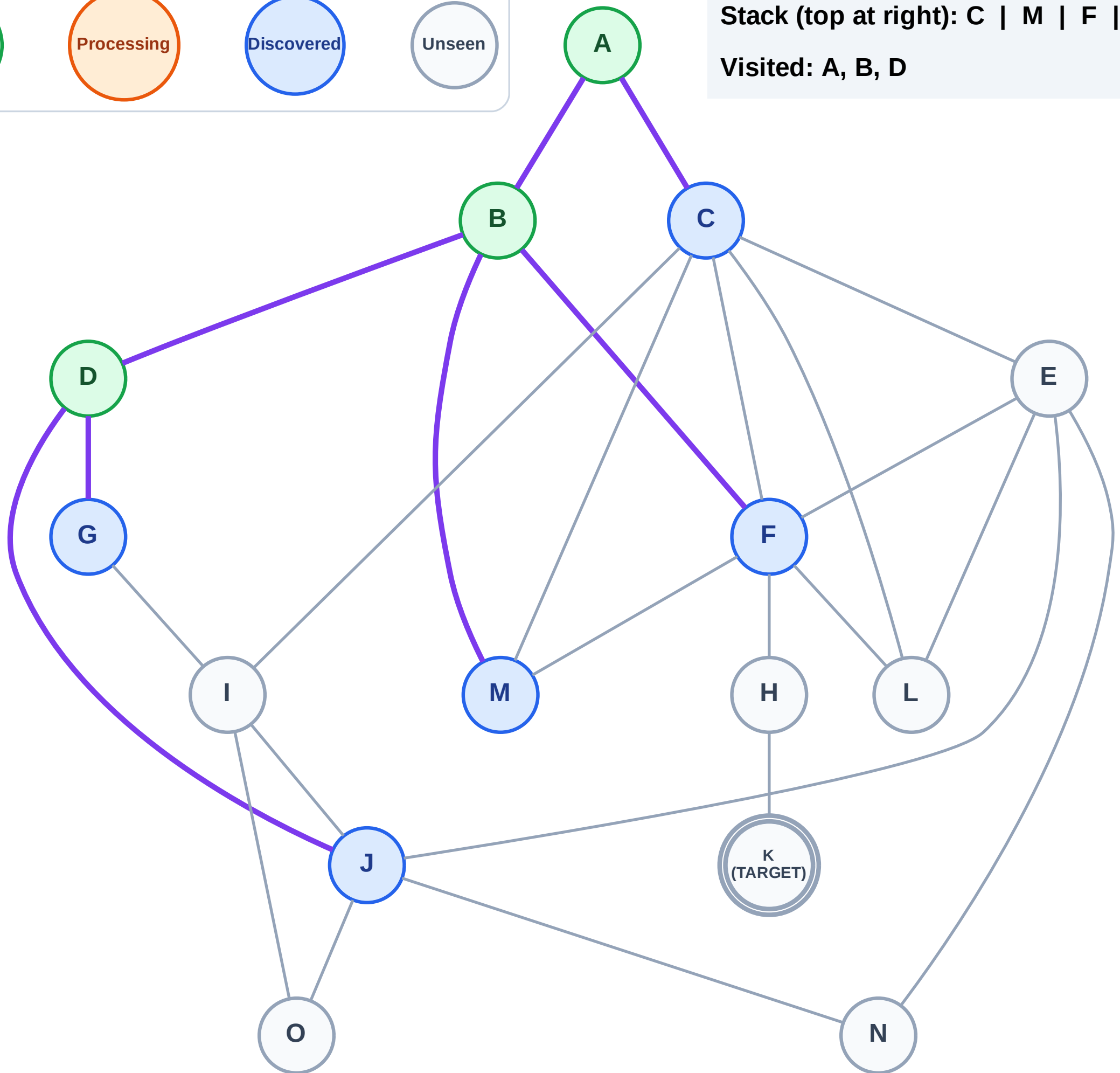
Step 7: Discover J, G from D

Legend



Stack (top at right): C | M | F | J | G

Visited: A, B, D



DFS Traversal

Step 8: Process node G

Legend

Complete

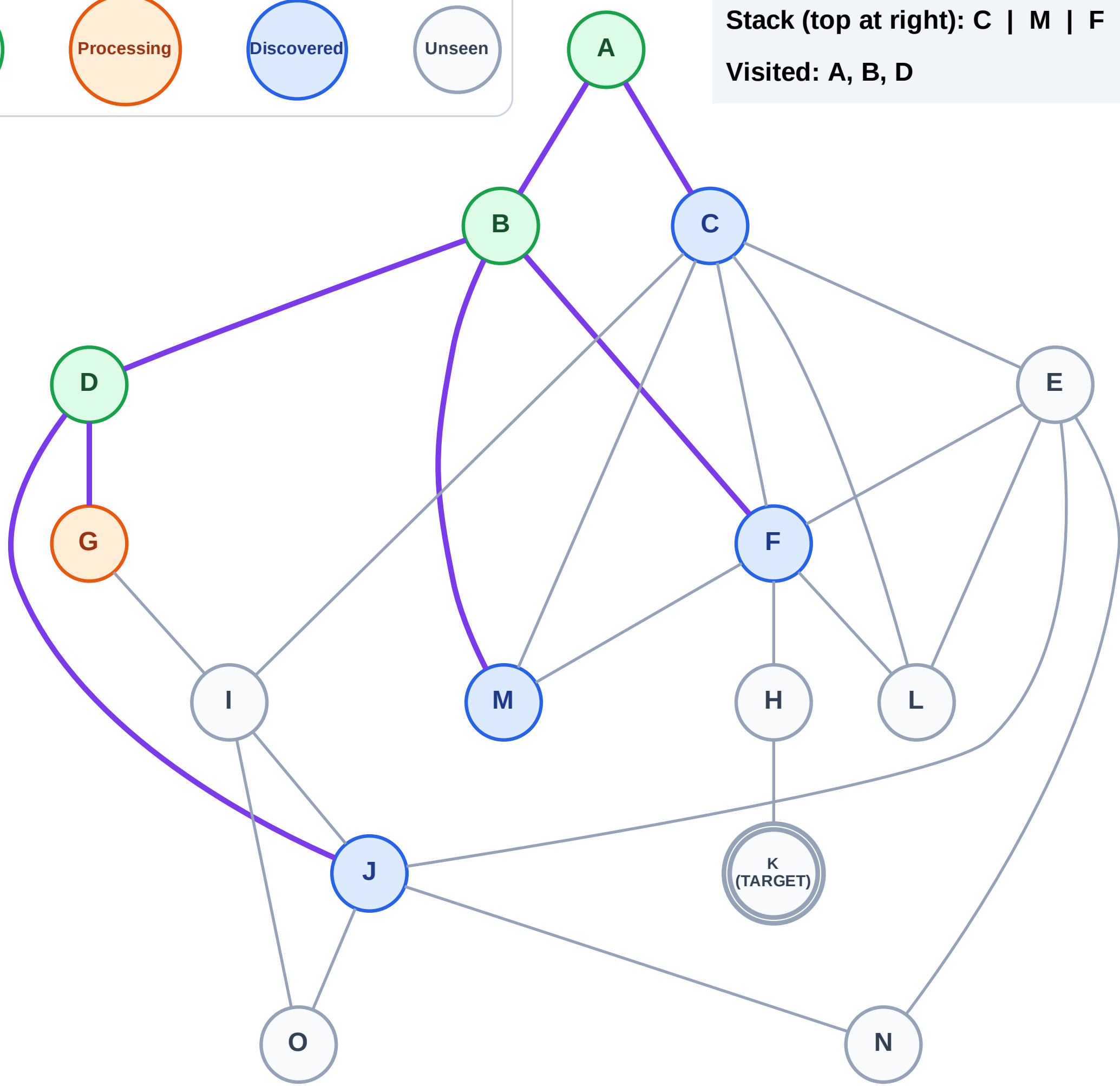
Processing

Discovered

Unseen

Stack (top at right): C | M | F | J

Visited: A, B, D



DFS Traversal

Step 9: Discover I from G

Legend

Complete

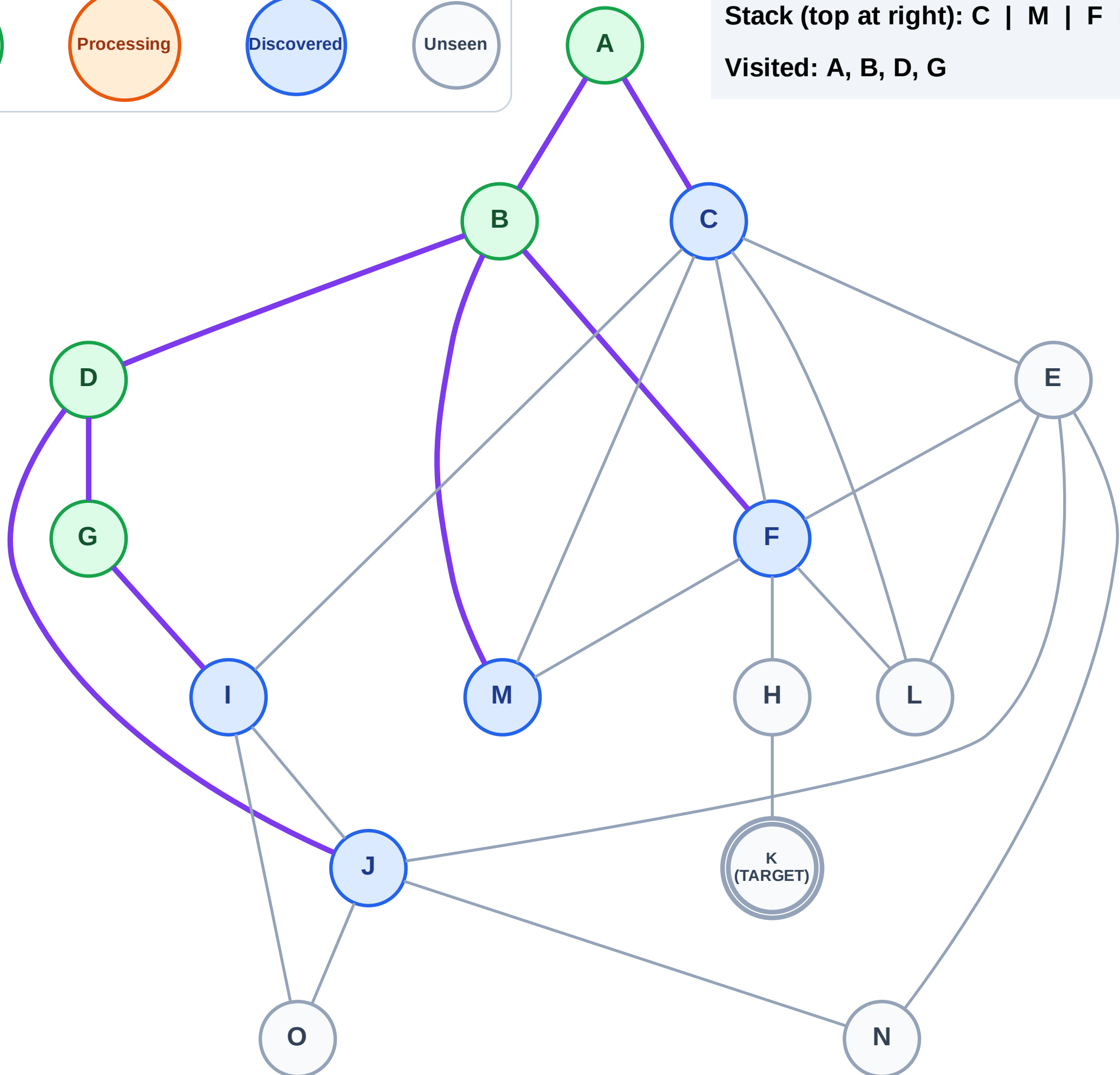
Processing

Discovered

Unseen

Stack (top at right): C | M | F | J | I

Visited: A, B, D, G



DFS Traversal

Step 10: Process node I

Legend

Complete

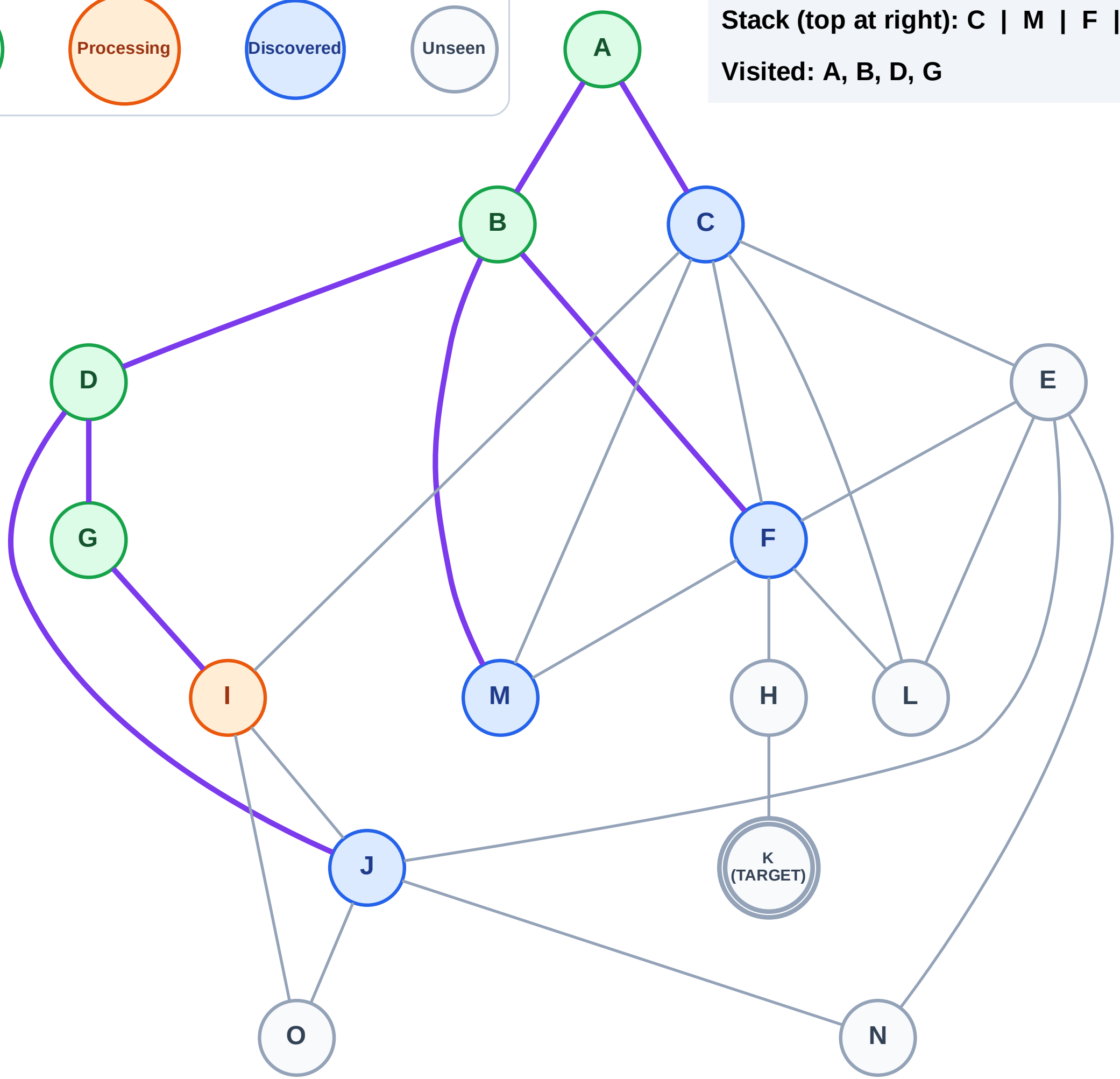
Processing

Discovered

Unseen

Stack (top at right): C | M | F | J

Visited: A, B, D, G



DFS Traversal

Step 11: Discover O from I

Legend

Complete

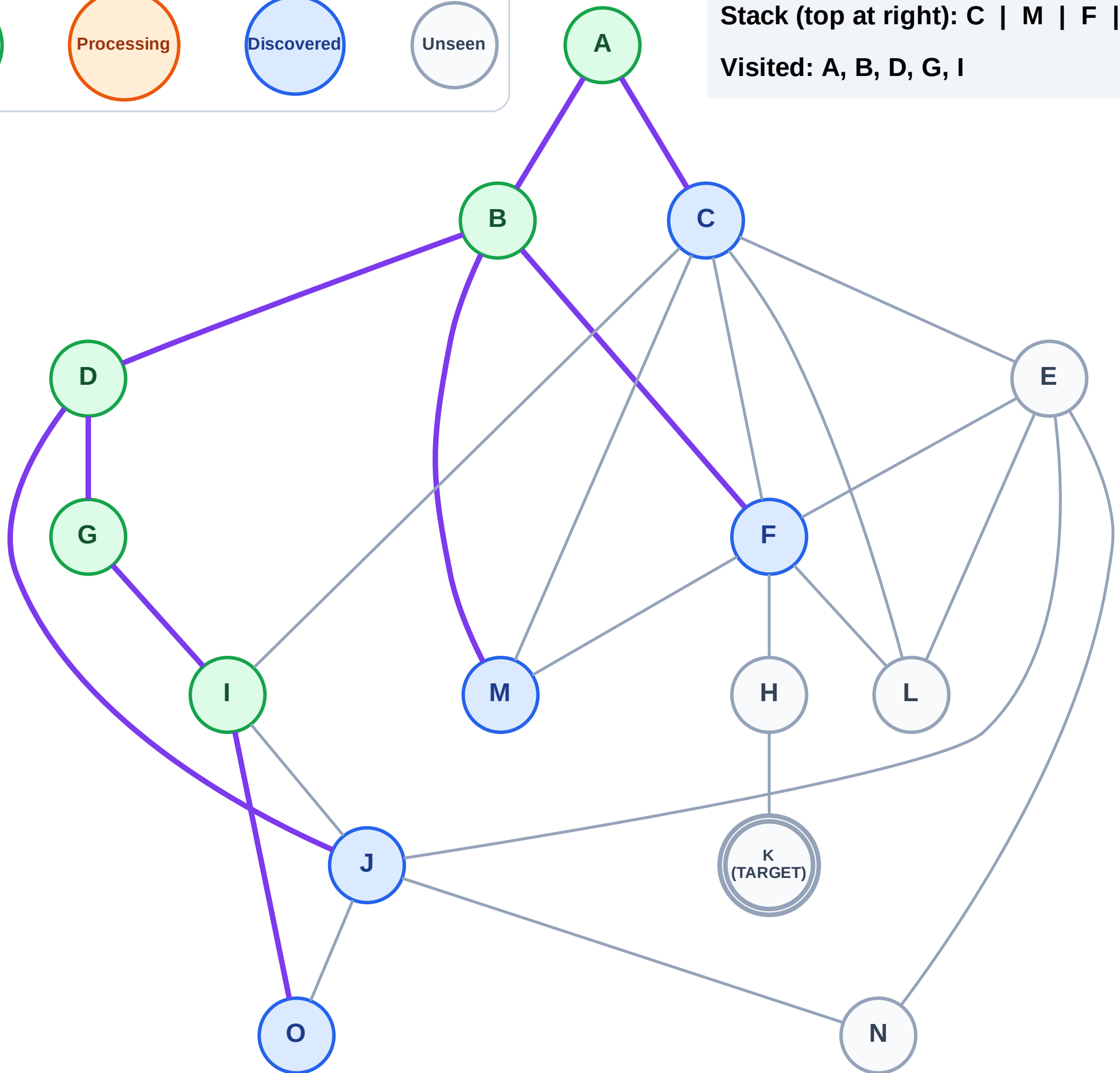
Processing

Discovered

Unseen

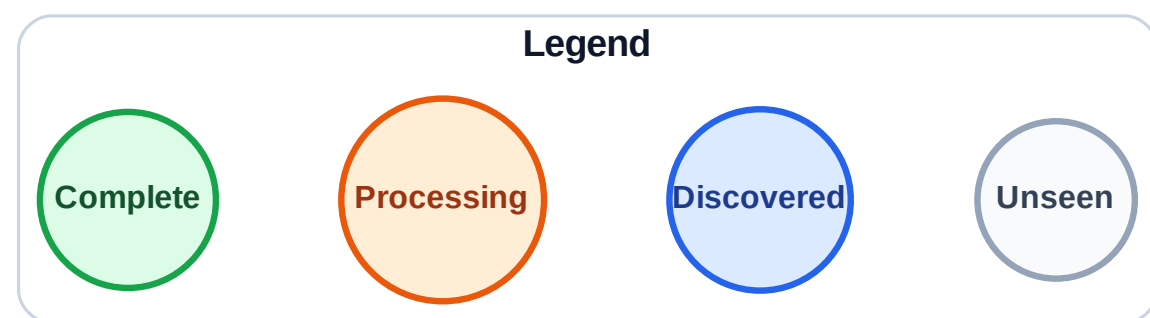
Stack (top at right): C | M | F | J | O

Visited: A, B, D, G, I

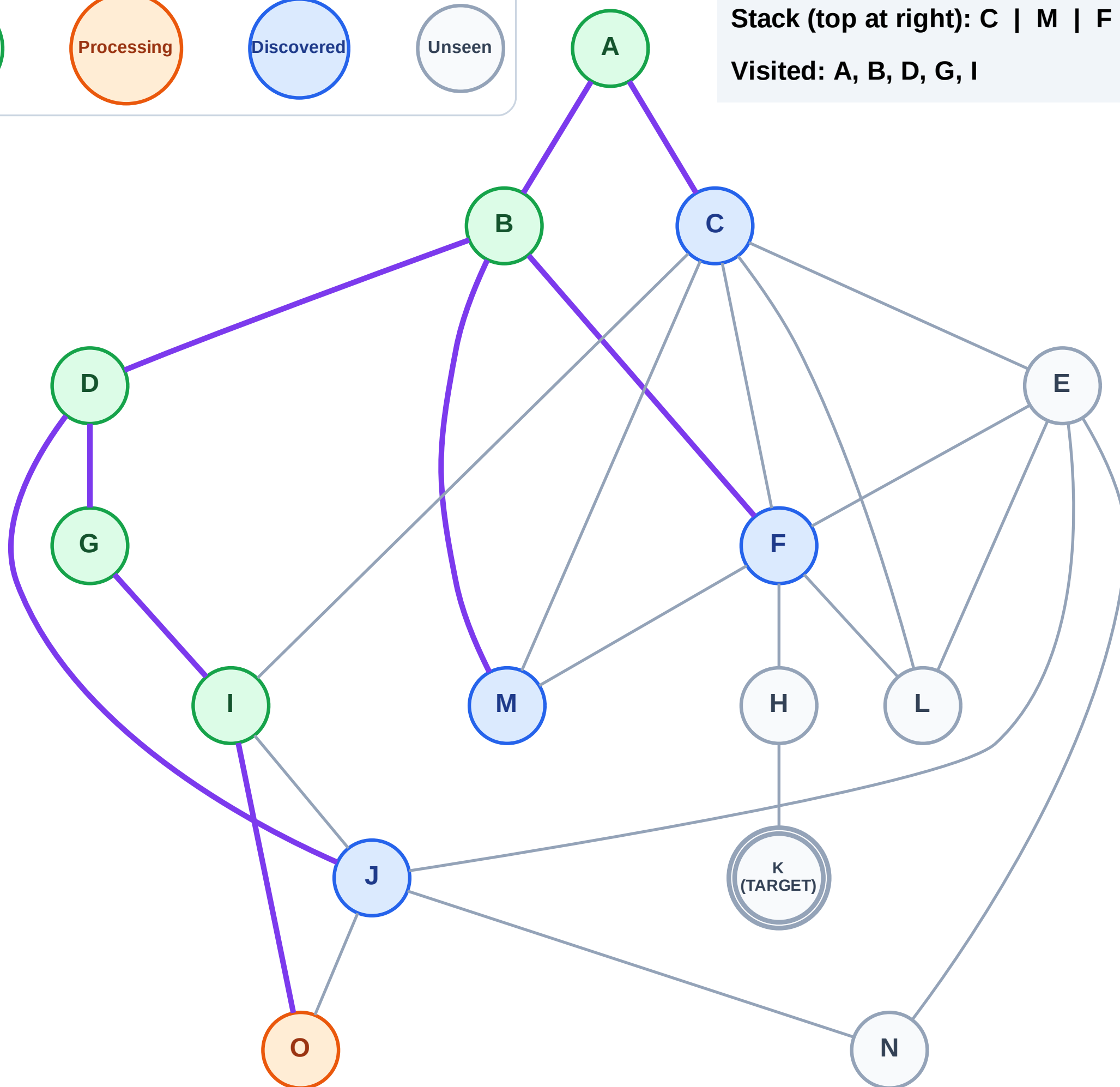


DFS Traversal

Step 12: Process node O



Stack (top at right): C | M | F | J
Visited: A, B, D, G, I



DFS Traversal

Step 13: No new neighbors from O

Legend

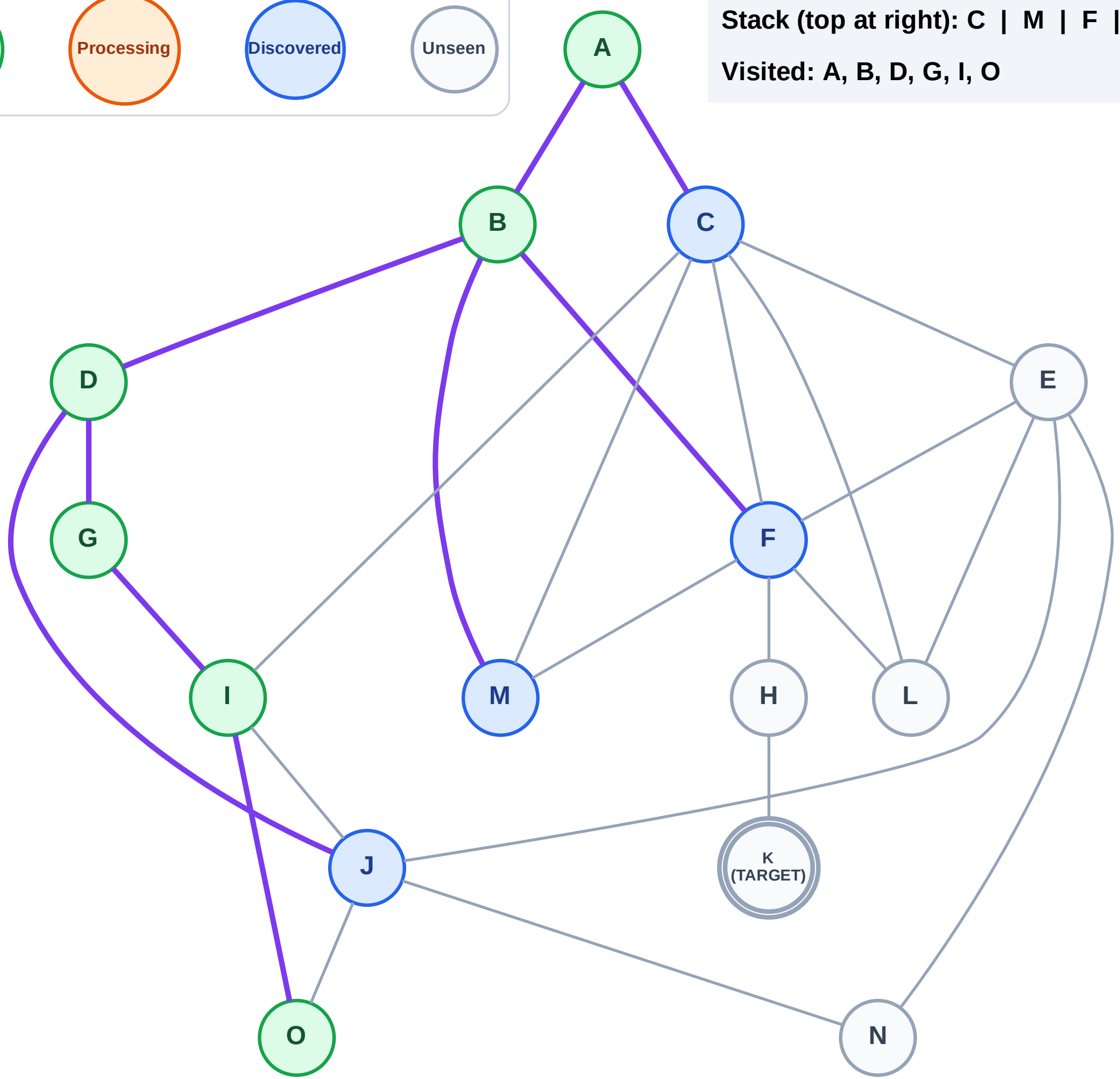
Complete

Processing

Discovered

Unseen

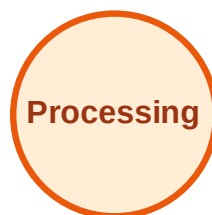
Stack (top at right): C | M | F | J
Visited: A, B, D, G, I, O



DFS Traversal

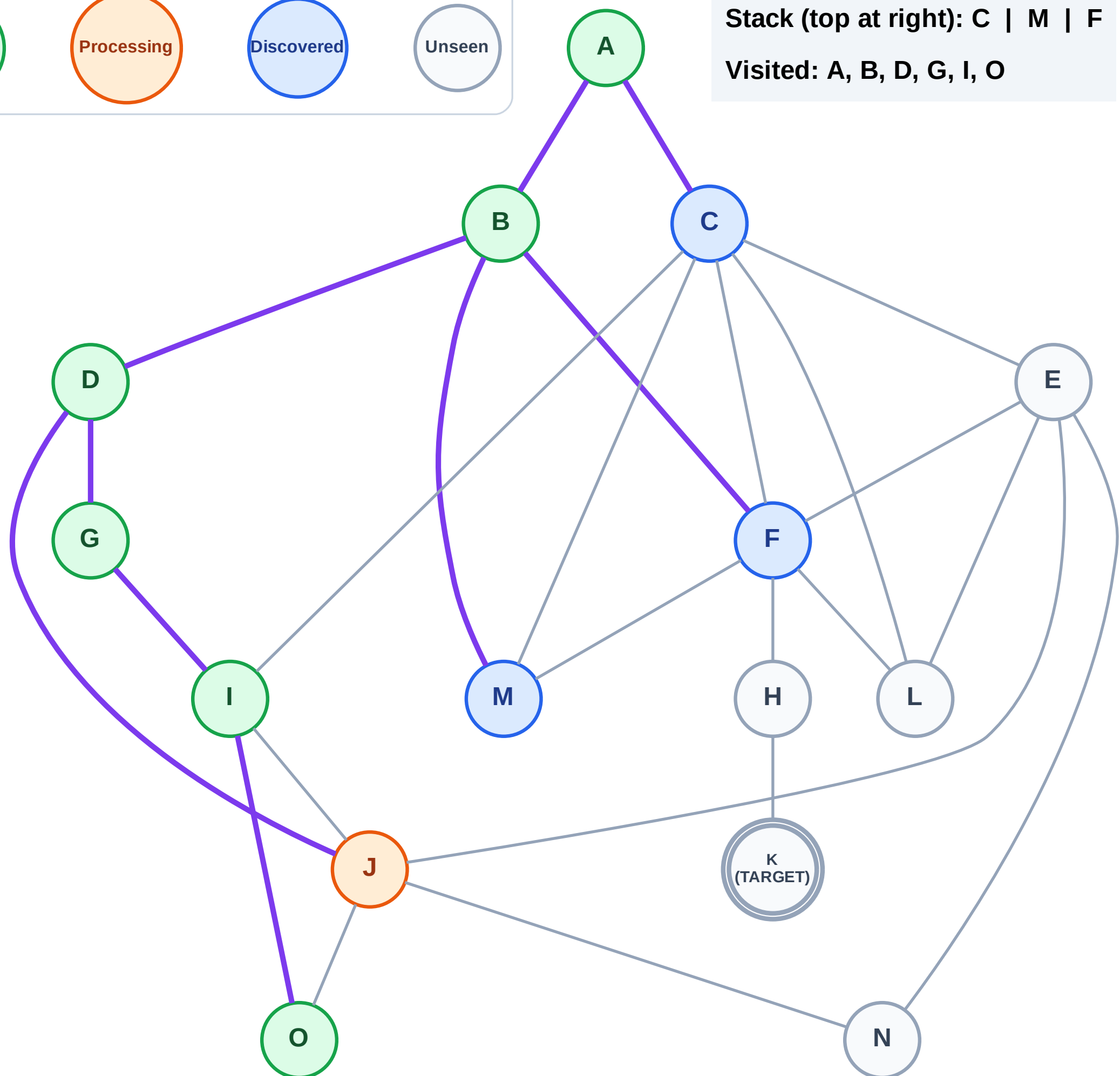
Step 14: Process node J

Legend



Stack (top at right): C | M | F

Visited: A, B, D, G, I, O

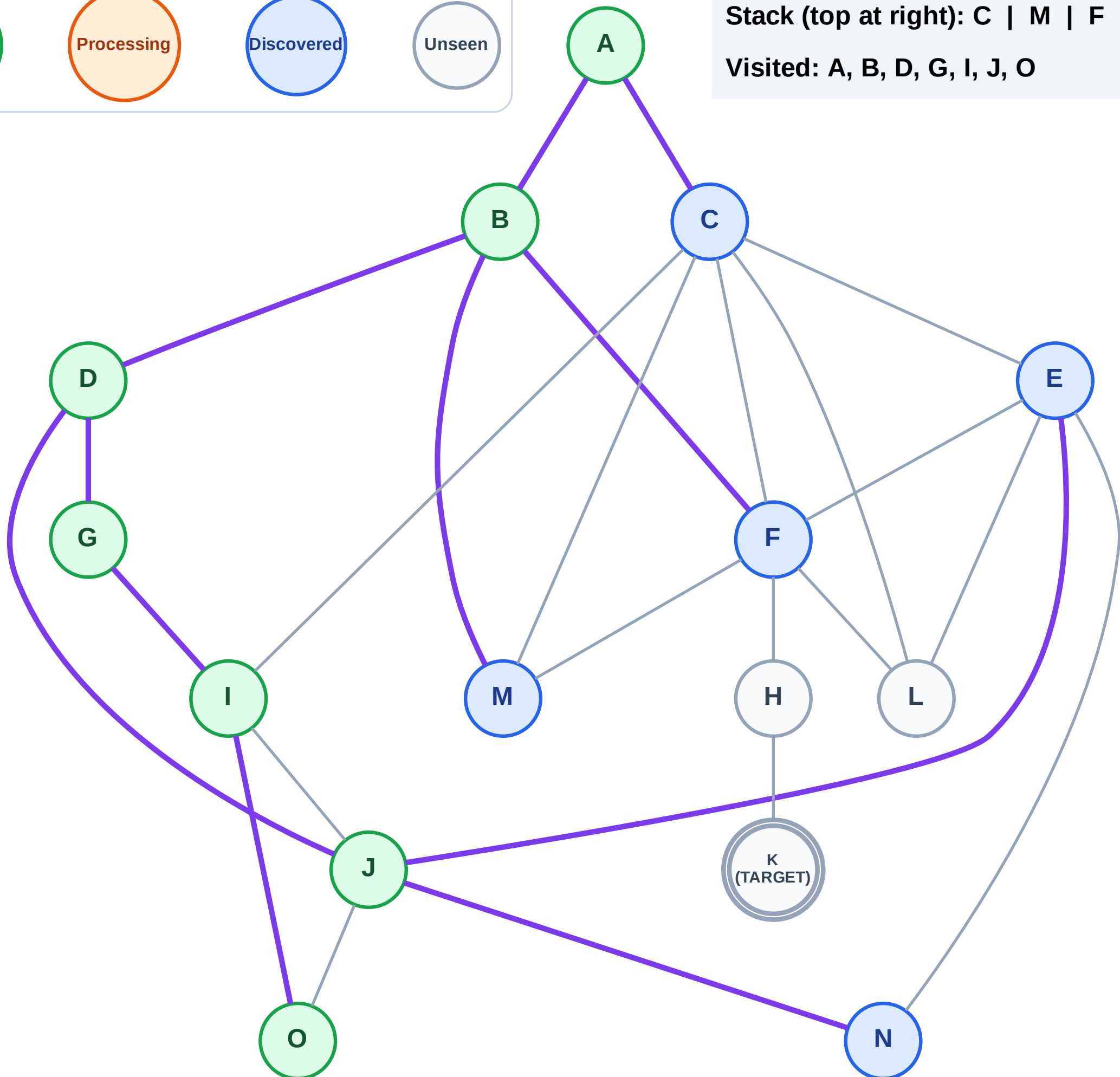


Step 15: Discover N, E from J

Legend

Unseen

Visited: A, B, D, G, I, J, O



DFS Traversal

Step 16: Process node E

Legend

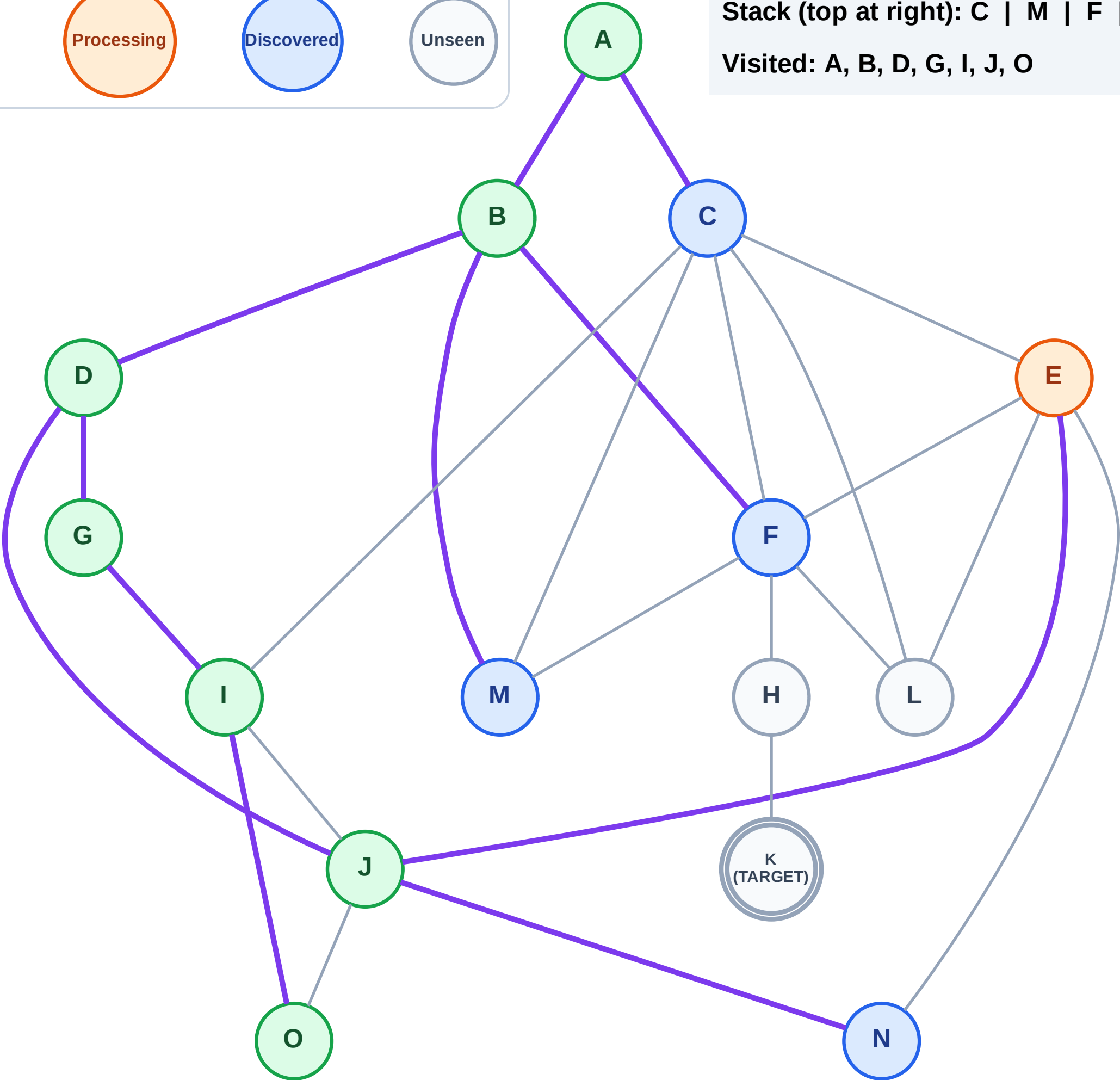
Complete

Processing

Discovered

Unseen

Stack (top at right): C | M | F | N
Visited: A, B, D, G, I, J, O



DFS Traversal

Step 17: Discover L from E

Legend

Complete

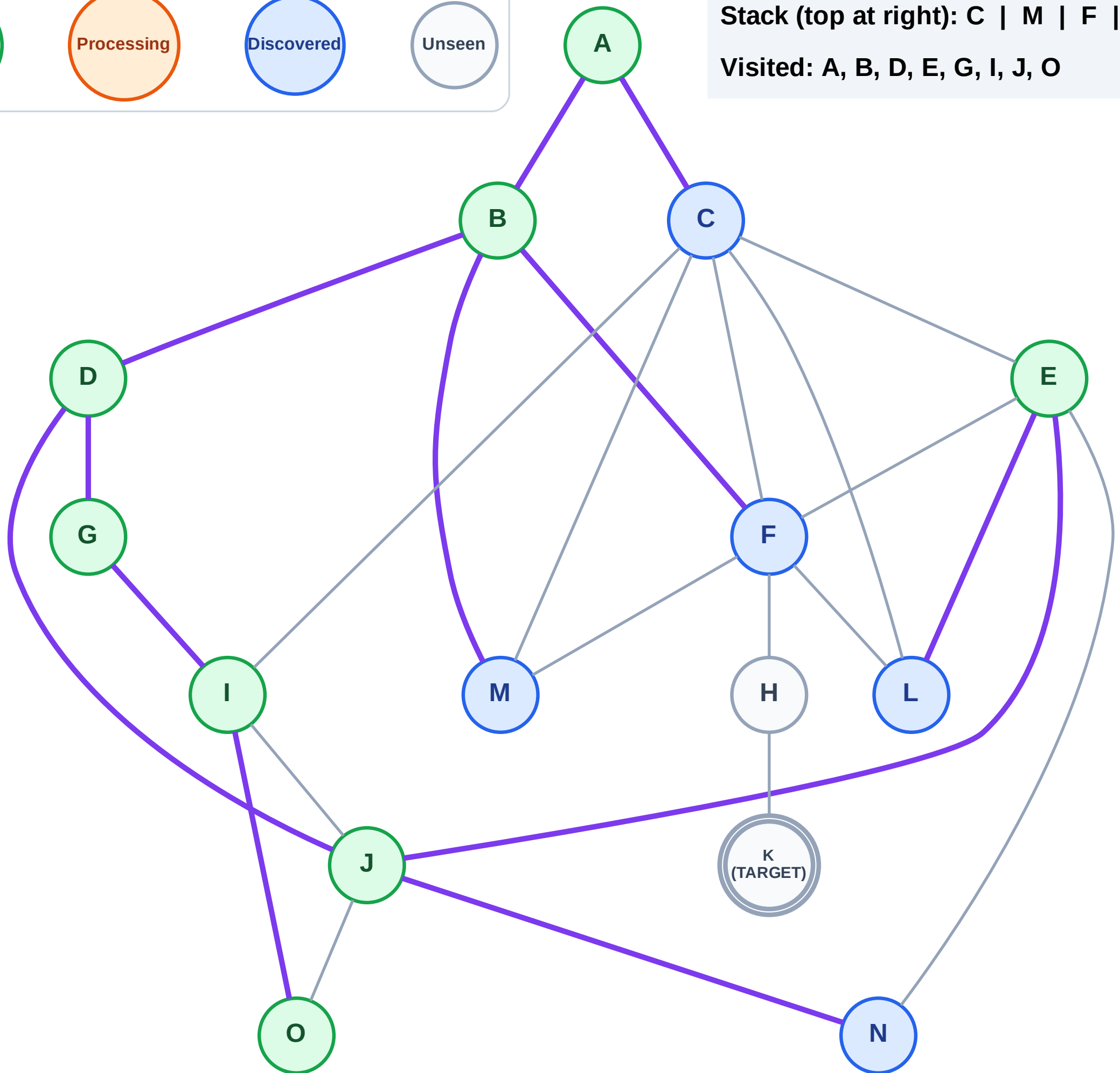
Processing

Discovered

Unseen

Stack (top at right): C | M | F | N | L

Visited: A, B, D, E, G, I, J, O



DFS Traversal

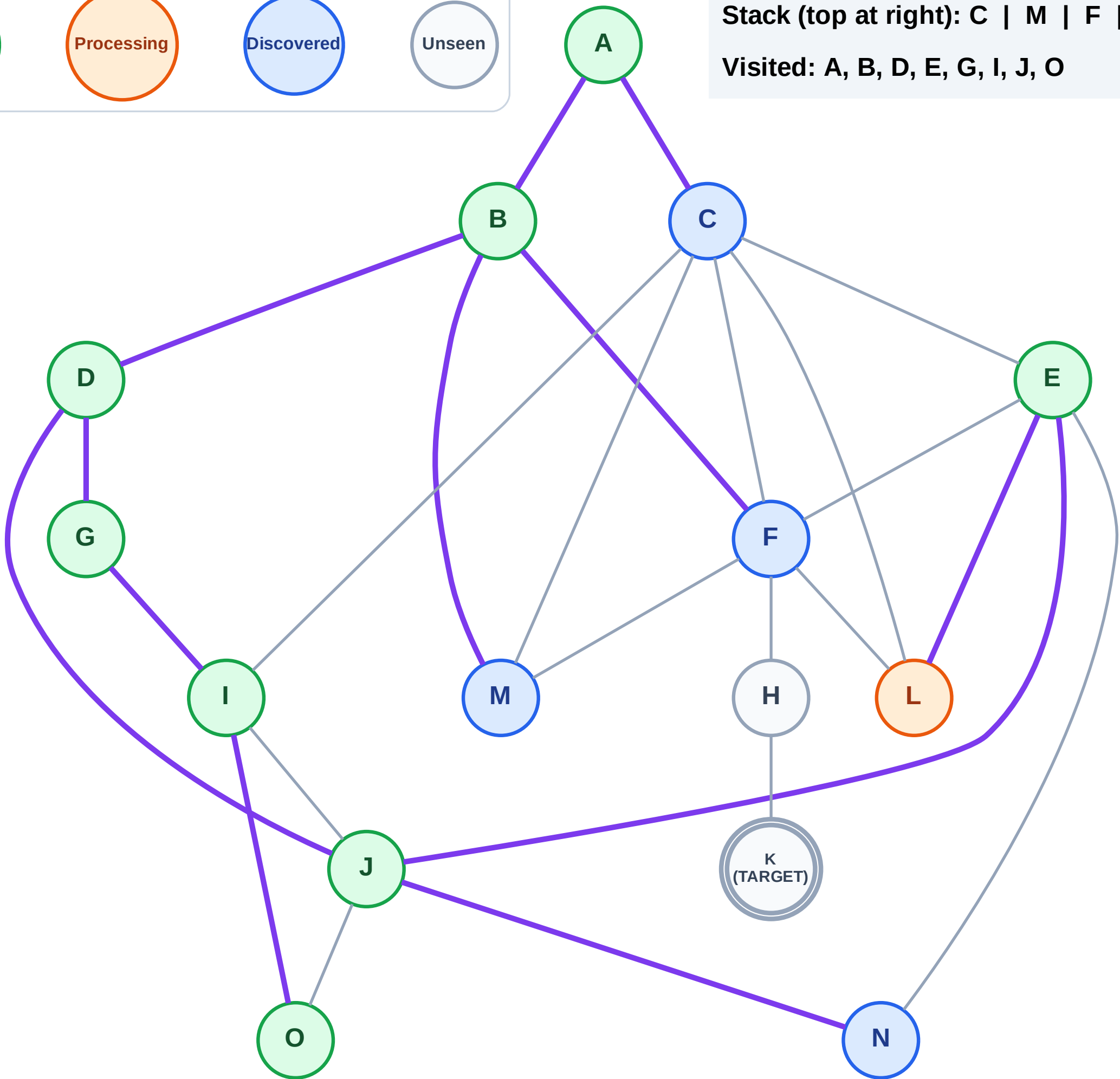
Step 18: Process node L

Legend



Stack (top at right): C | M | F | N

Visited: A, B, D, E, G, I, J, O



DFS Traversal

Step 19: No new neighbors from L

Legend

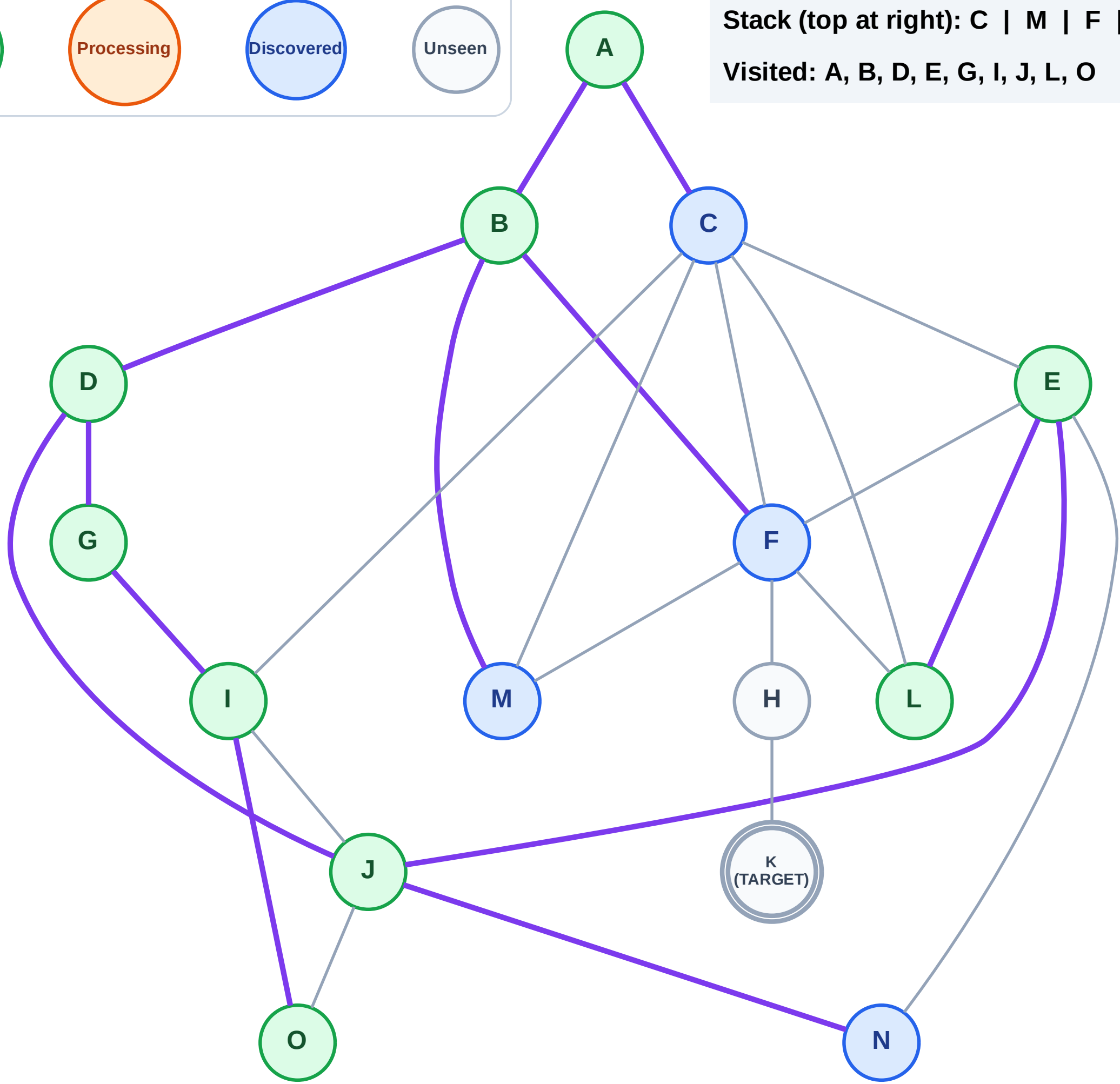
Complete

Processing

Discovered

Unseen

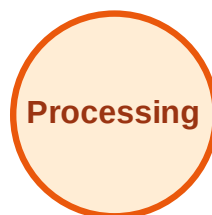
Stack (top at right): C | M | F | N
Visited: A, B, D, E, G, I, J, L, O



DFS Traversal

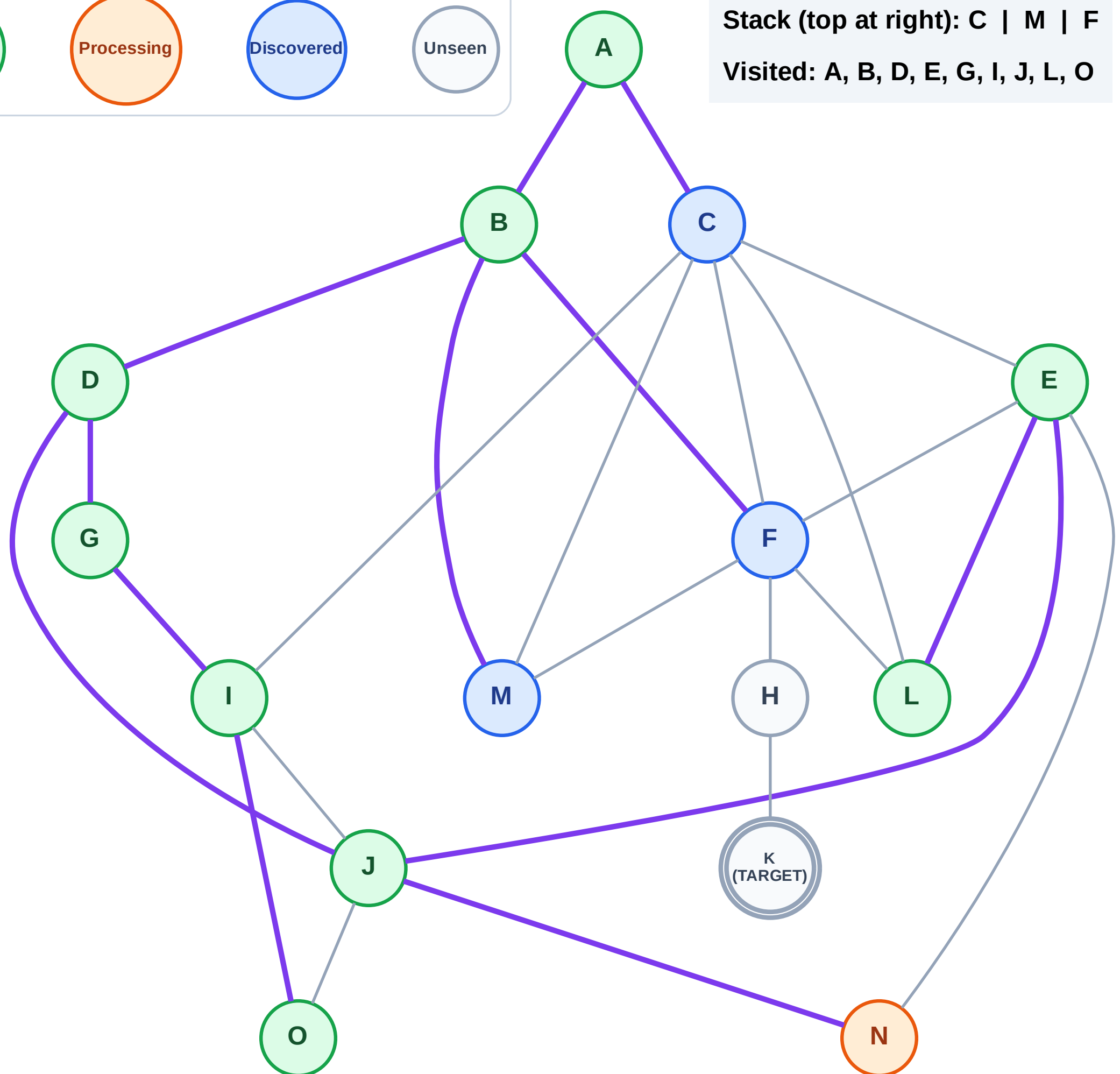
Step 20: Process node N

Legend



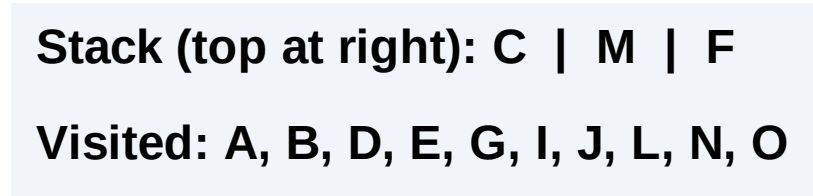
Stack (top at right): C | M | F

Visited: A, B, D, E, G, I, J, L, O



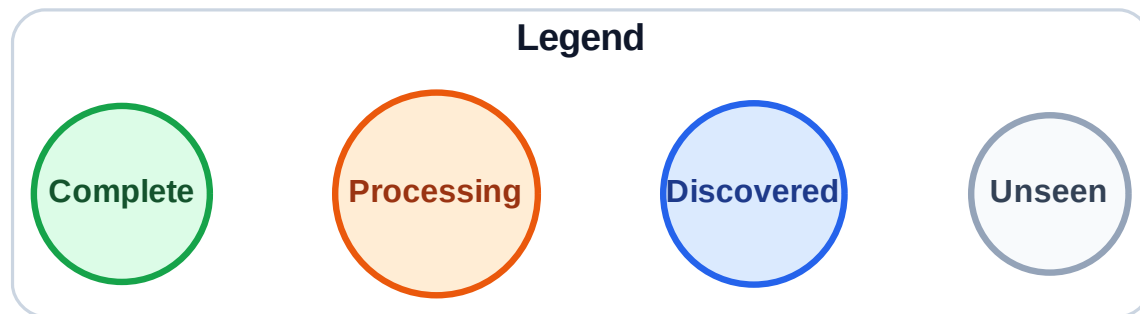
Step 21: No new neighbors from N

Step 21: No new neighbors from N



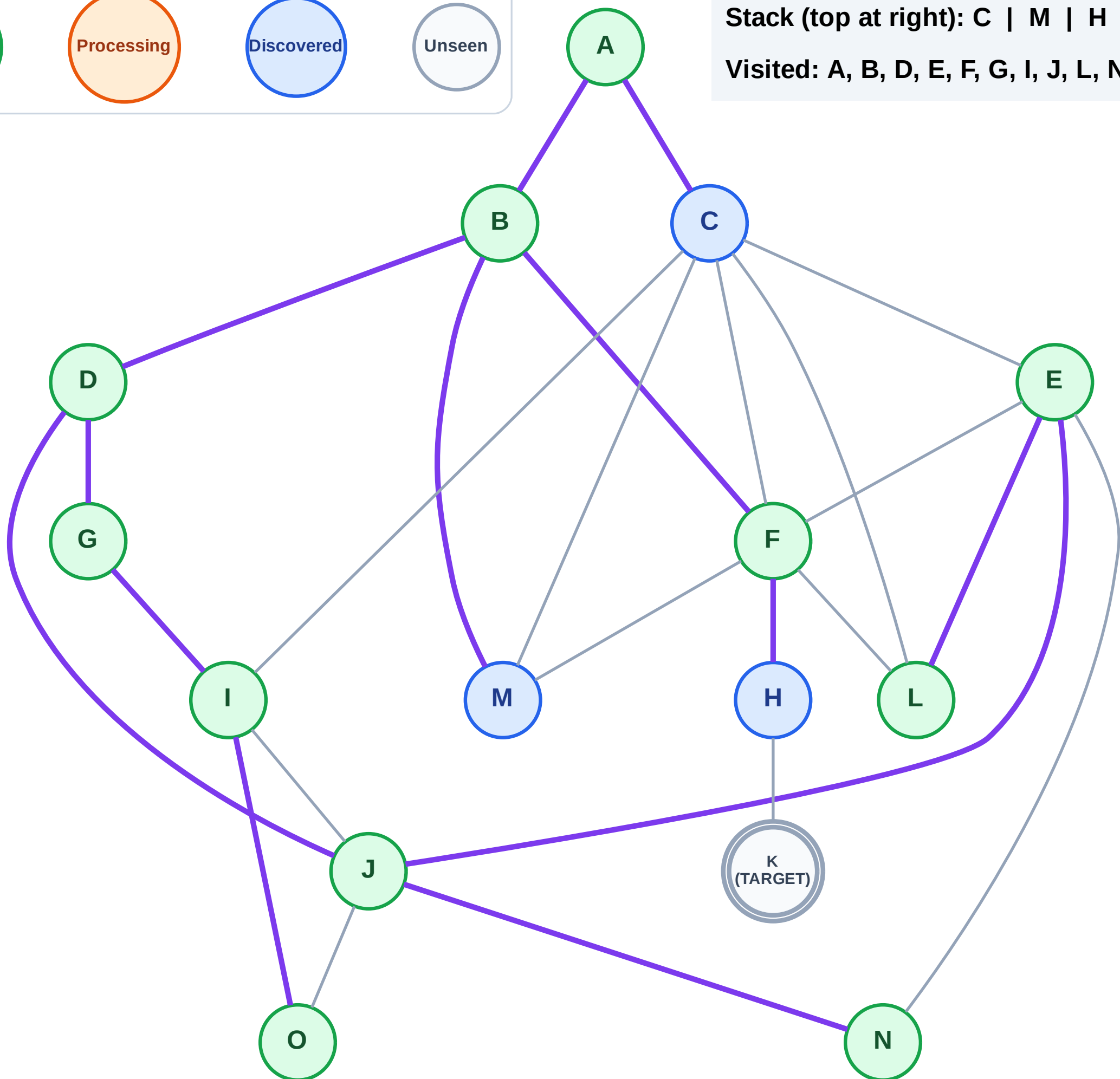
Step 23: Discover H from F

Step 23: Discover H from F



Stack (top at right): C | M | H

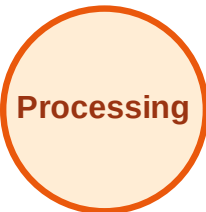
Visited: A, B, D, E, F, G, I, J, L, N, O



DFS Traversal

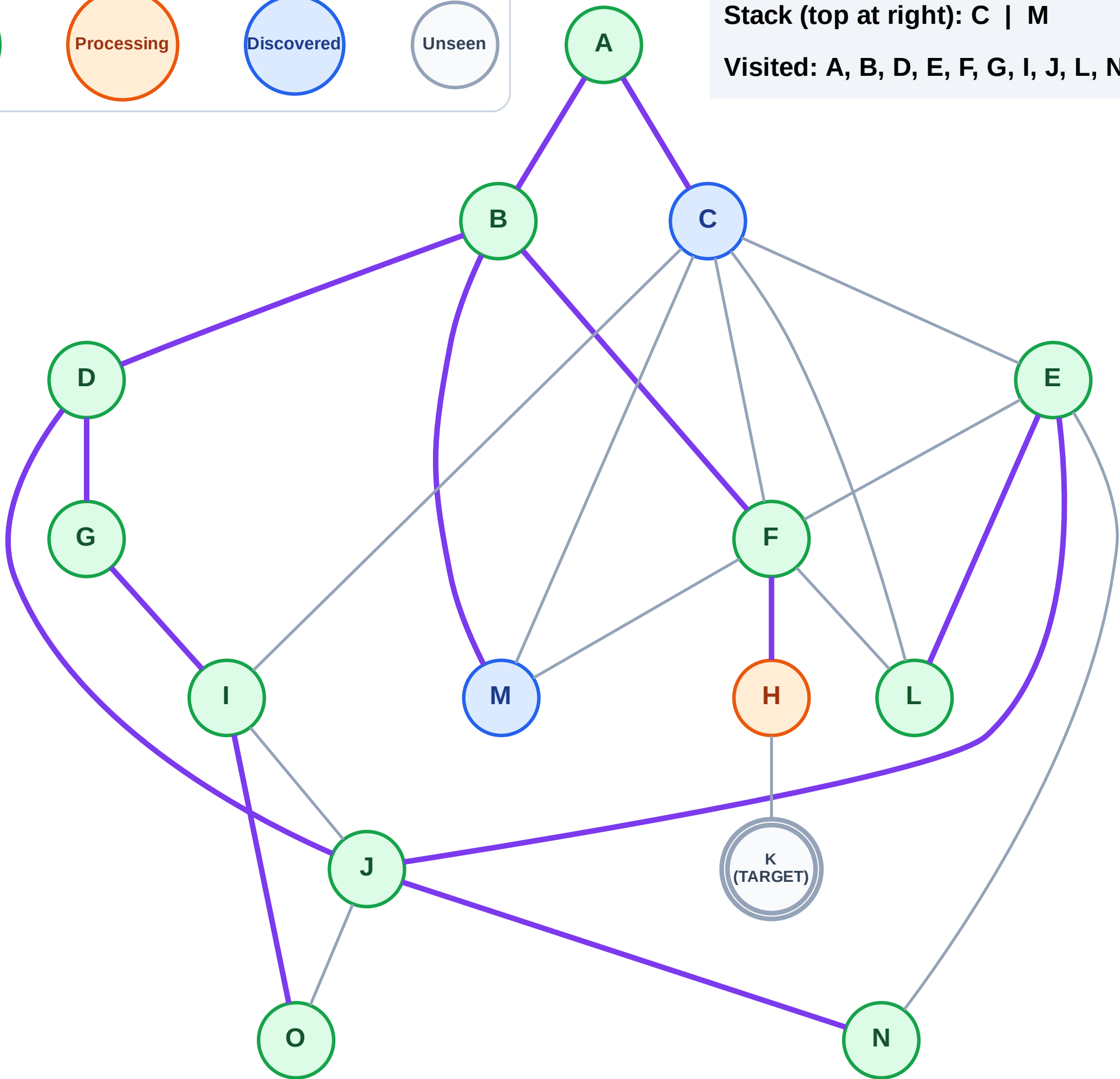
Step 24: Process node H

Legend



Stack (top at right): C | M

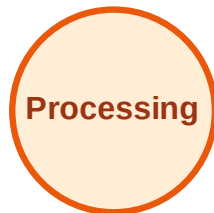
Visited: A, B, D, E, F, G, I, J, L, N, O



DFS Traversal

Step 25: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → F → H → K

Legend



Stack (top at right): C | M

Visited: A, B, D, E, F, G, H, I, J, K, L, N, O

